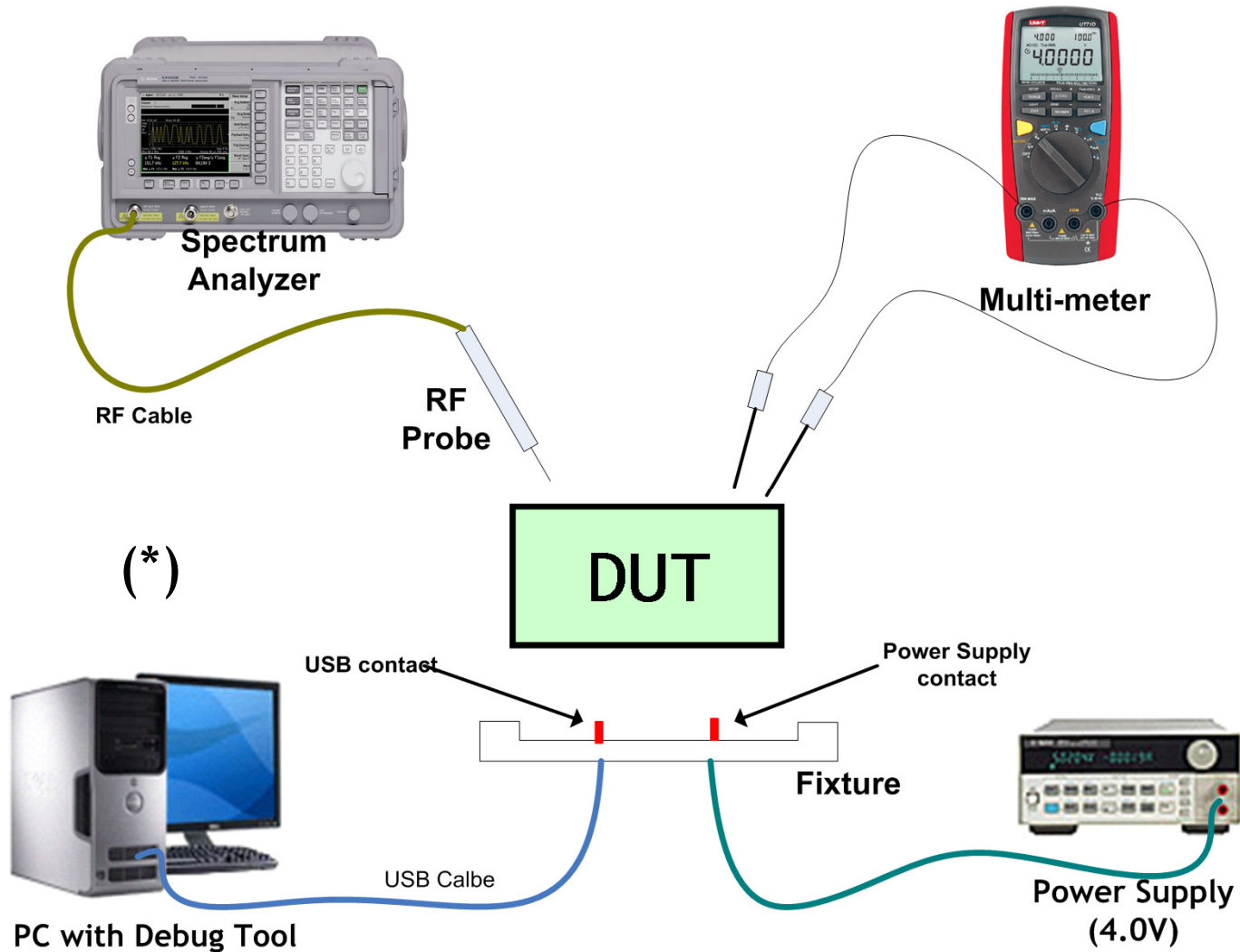


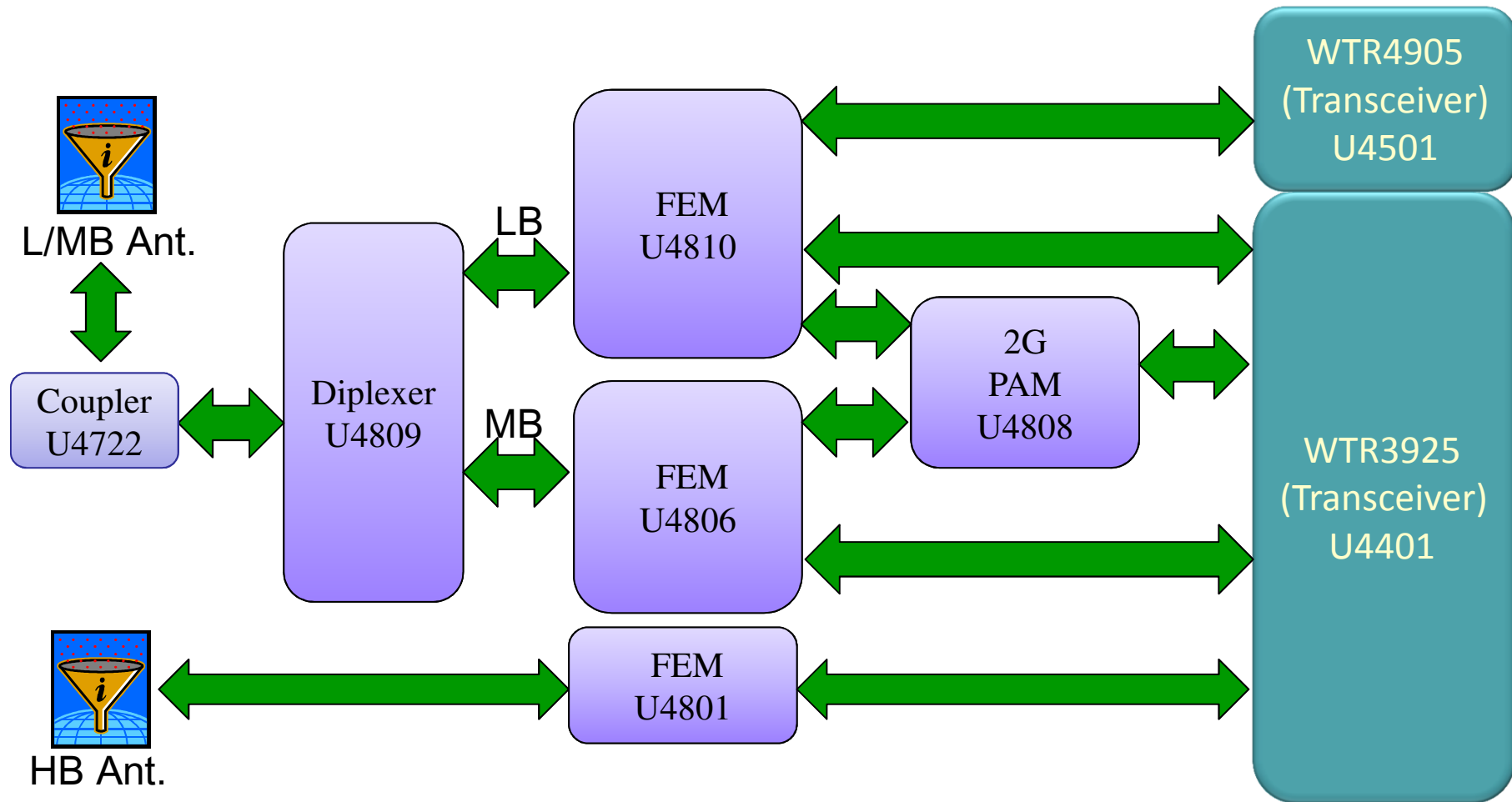
ASUS ZS570KL

RF Hardware Trouble Shooting Guide

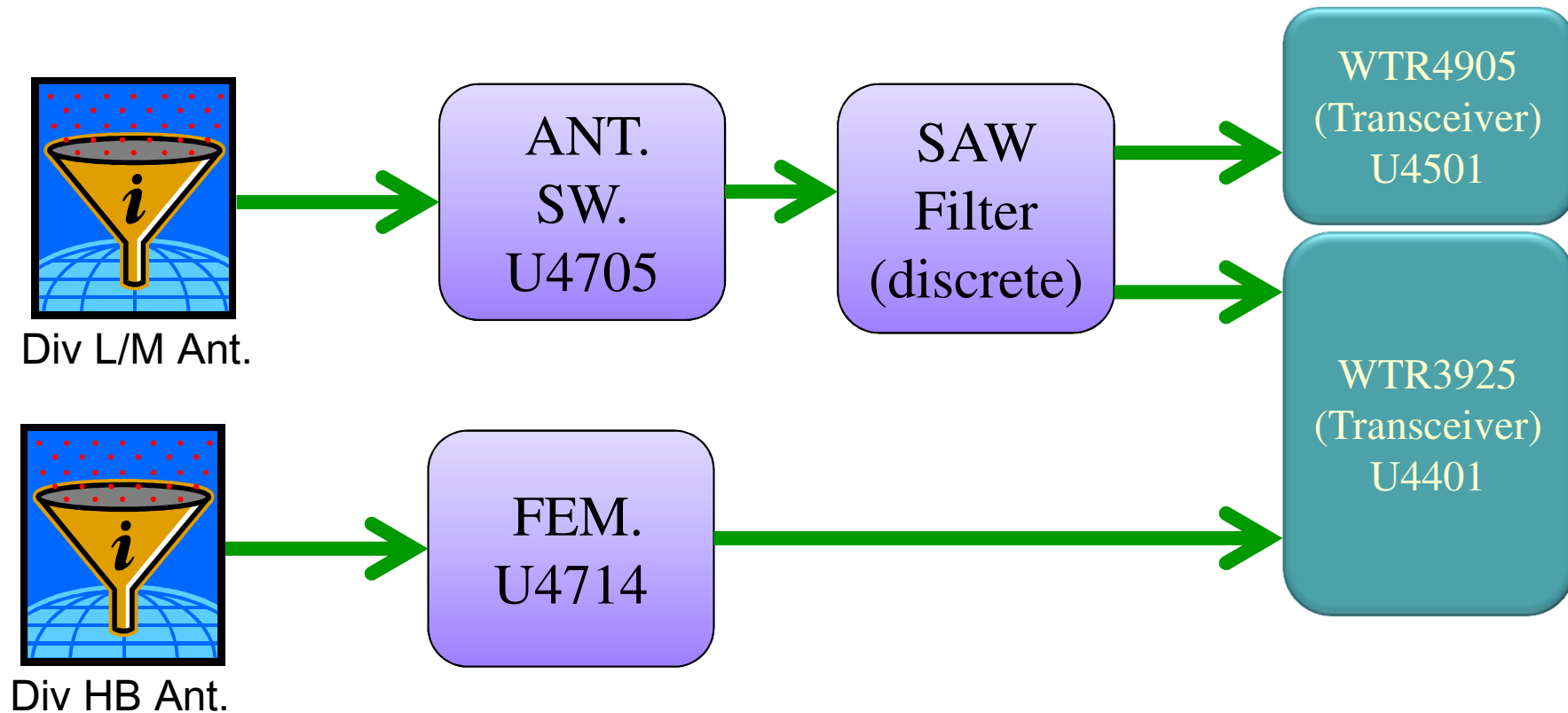
B/L Repair and Test Environment



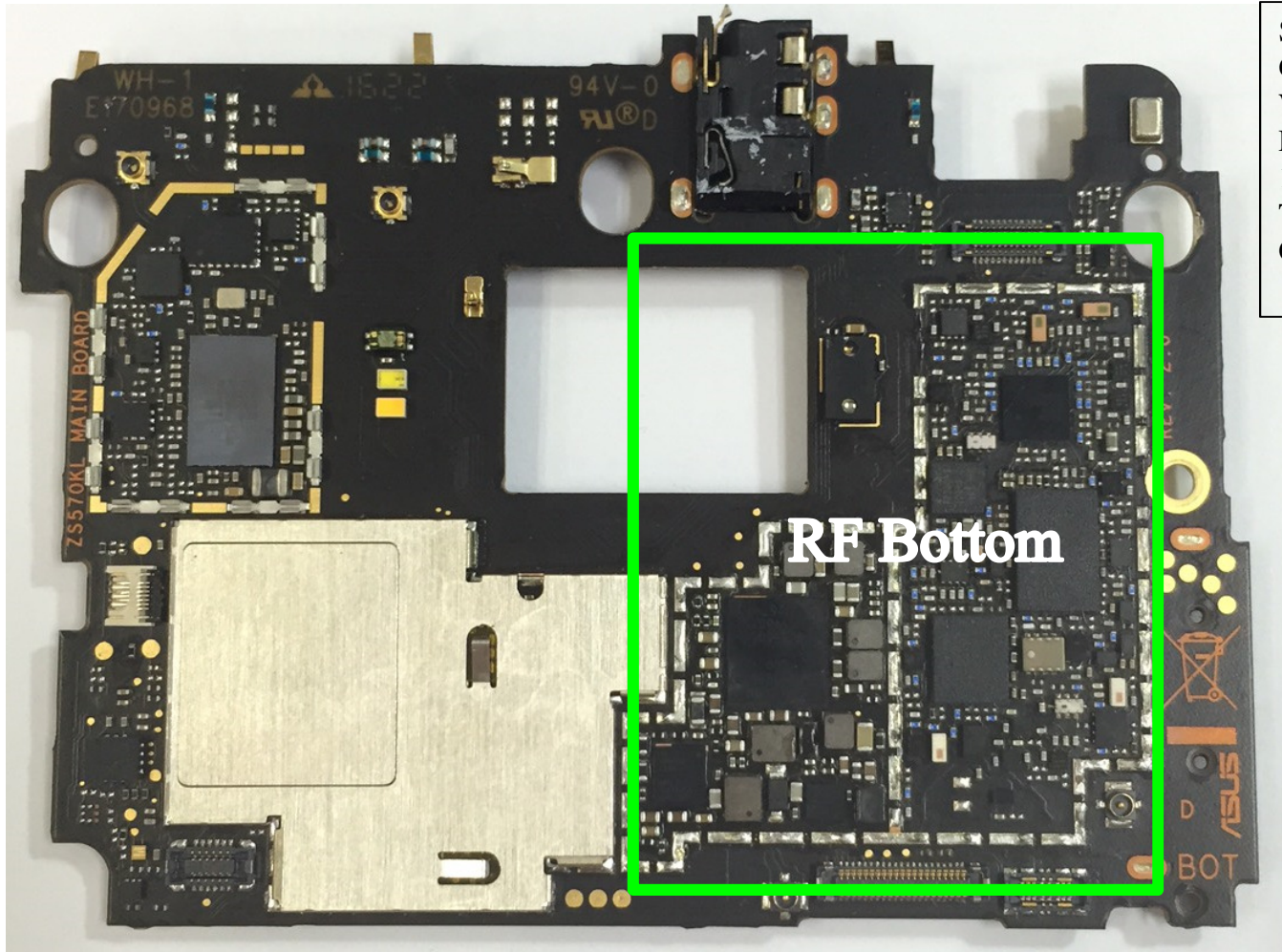
2G/3G/4G Main Antenna Block Diagram



3G/4G Diversity Antenna Block Diagram



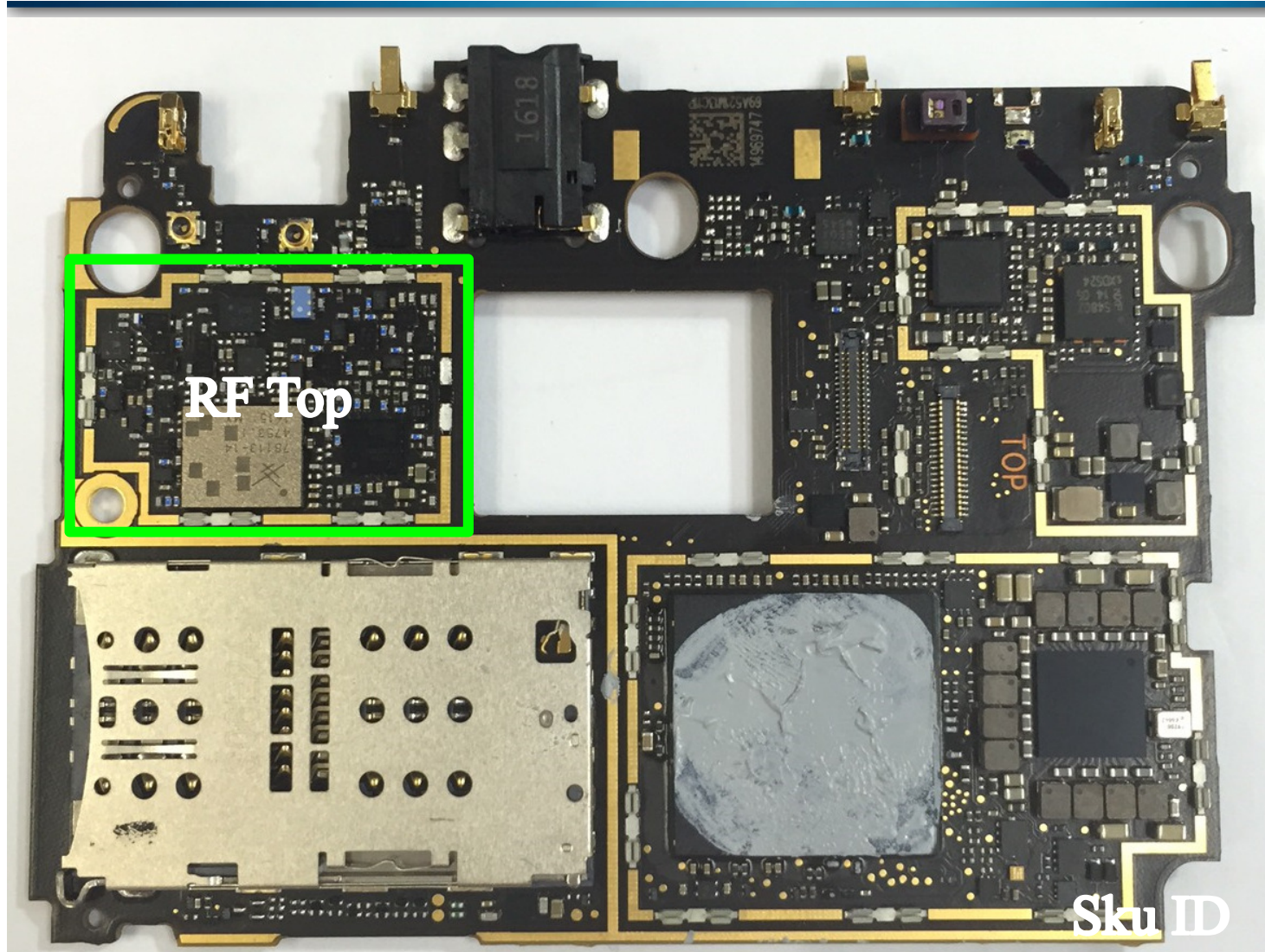
ZS570KL RF Portion From PCBA Bottom View



Supported Bands
GSM: 850/900/DCS/PCS
WCDMA: Band 1/2/4/5/8/34/39
FDD-LTE Band
1/2/3/4/5/7/8/12/17/18/19/20/26/28/29/30
TDD-LTE: 38/39/40/41
CDMA2000: B0

Low/Mid/High Band Definition
LB: 5 8 12 17 18 19 20 26 28 29 B0
GSM850 GSM900
MB: 1 2 3 4 34 39 DCS PCS
HB: 7 30 38 40 41

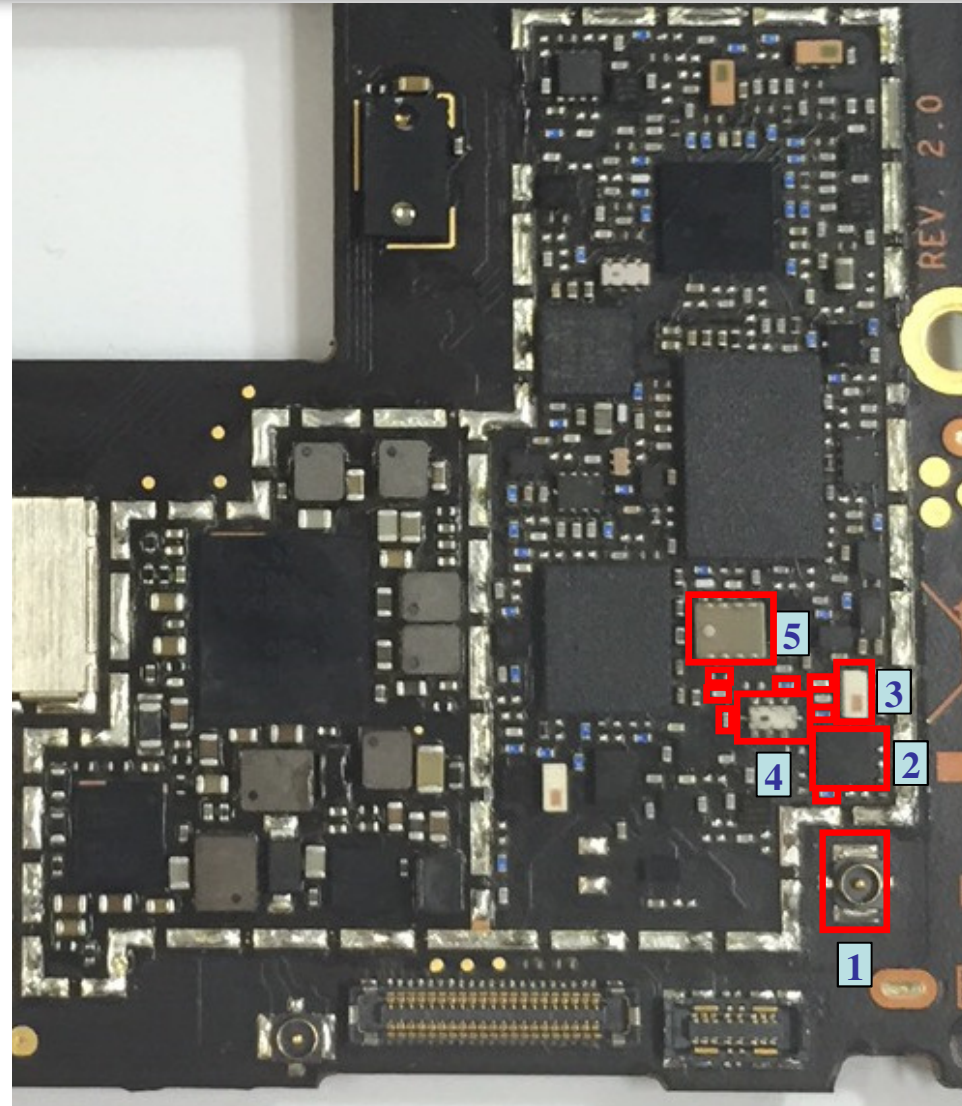
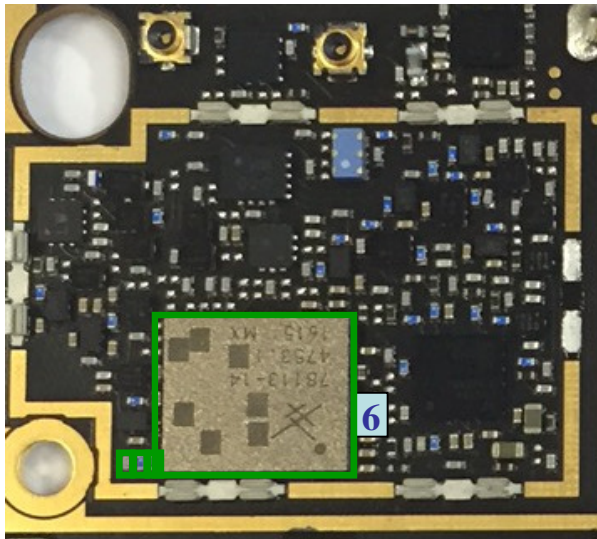
ZS570KL RF Portion From PCBA Top View



LB TRx Common Comp.

1. CON4701: Connector
2. U4708: DPDT
3. U4702: LPF
4. U4722: Coupler
5. U4809: Diplexer
6. U4810: LB FEM

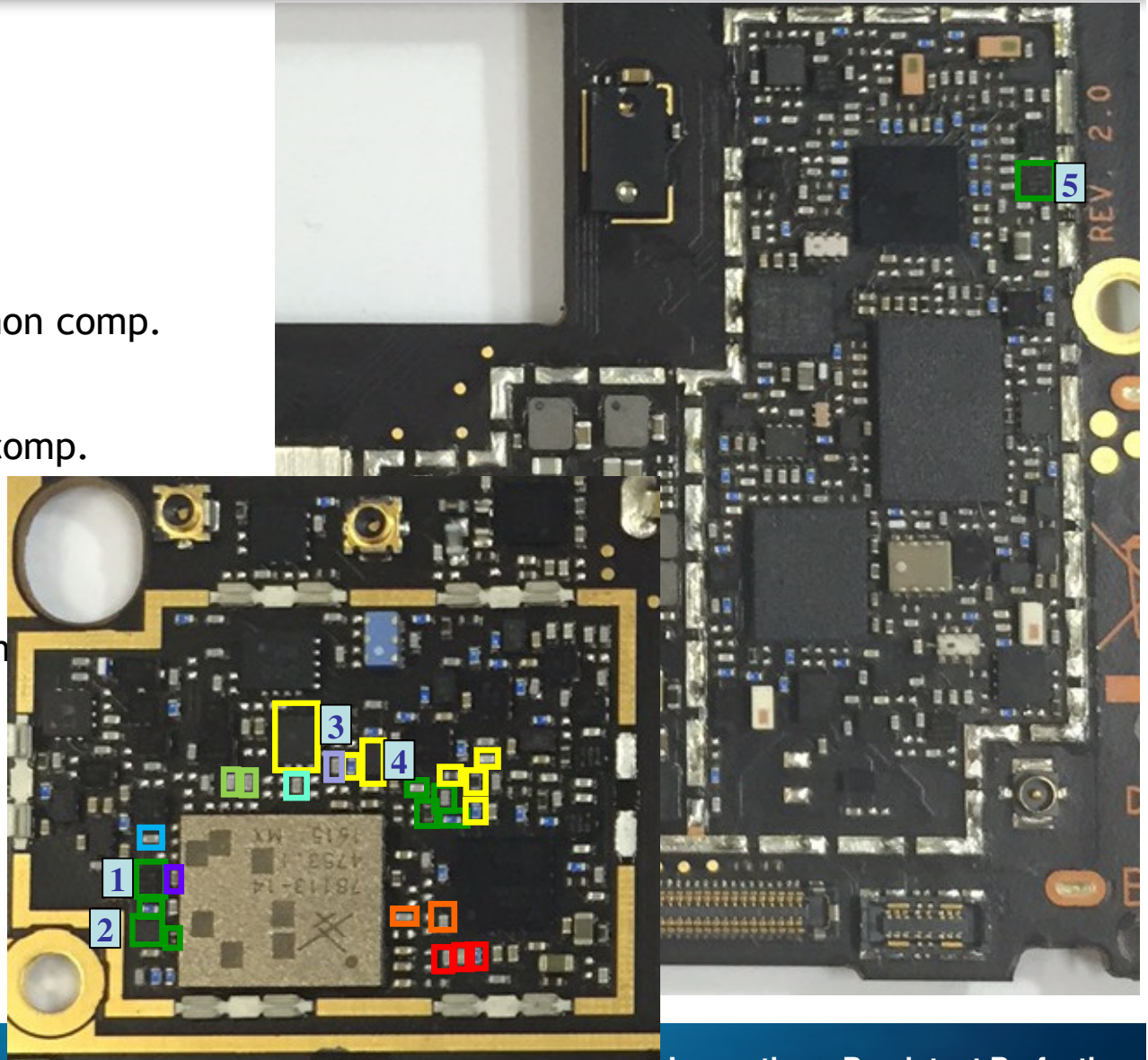
 Top
 Bottom



LB TRx Comp.

1. U4915: SPDT
2. U4912: LNA
3. U4917: SP4T
4. U4918: LNA
5. U4913: SPDT

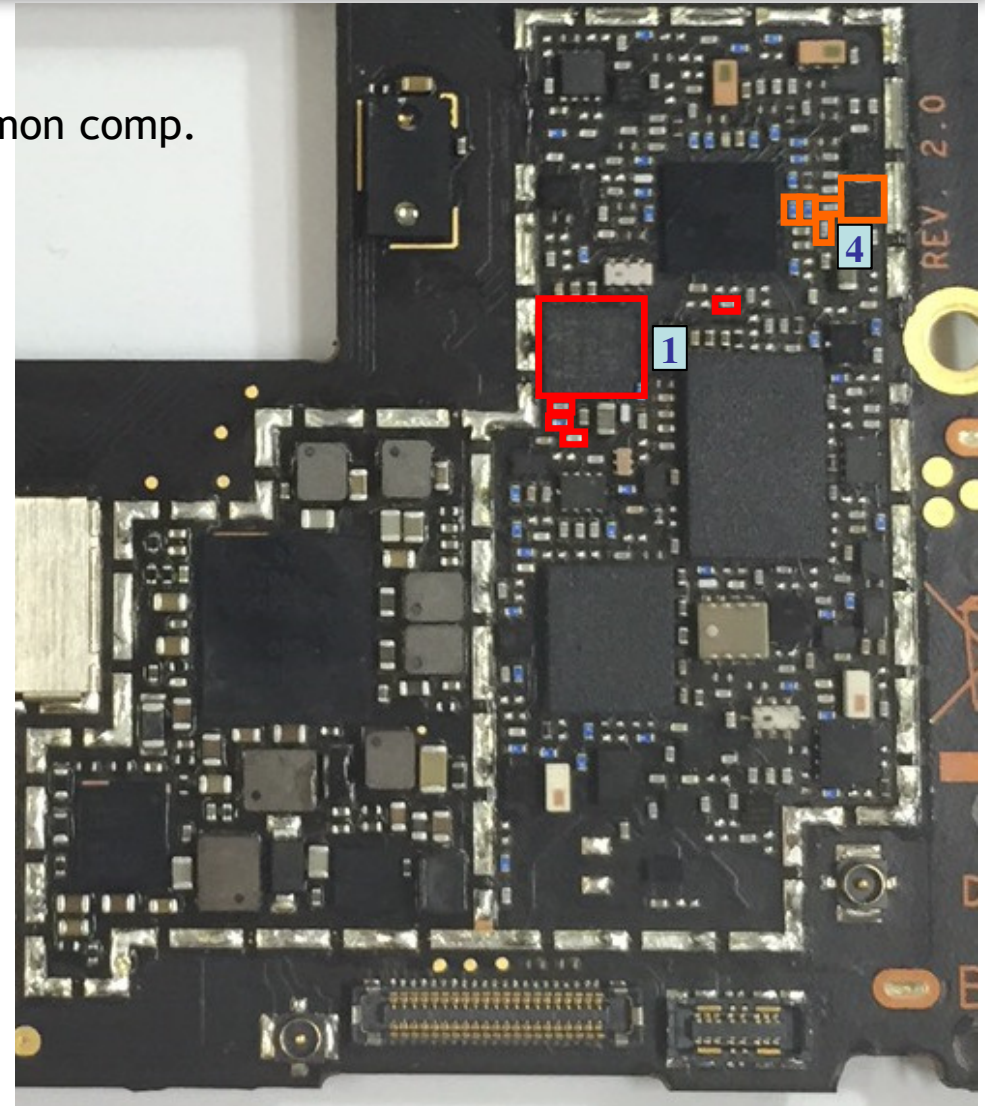
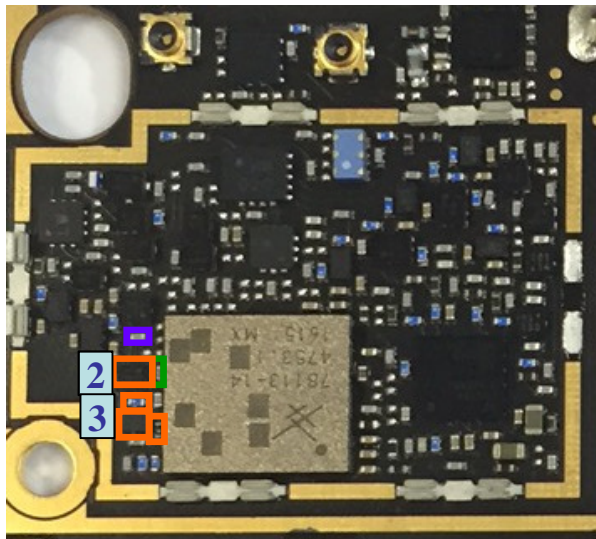
- B5 B8 B19 B20 B28 Tx common comp.
- B12 B17 Tx common comp.
- B5 B8 B19 B26 Rx common comp.
- B5 Rx comp.
- B8 Rx comp.
- B12 B17 B20 B28 B29 Rx com
- B12 B17 Rx comp.
- B20 Rx comp.
- B28 Rx comp.
- B29 Rx comp.



GSM 850/900 TRx (LB)

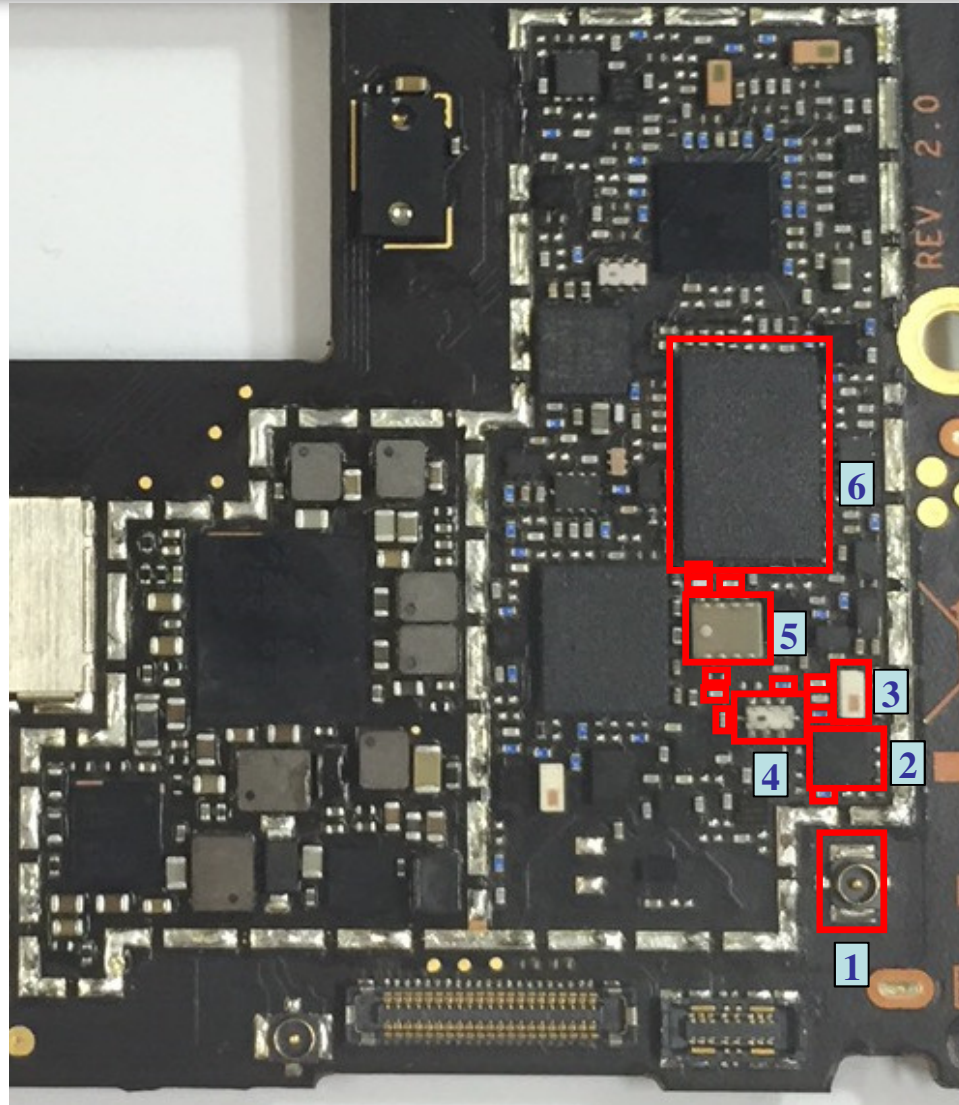
1. U4808: 2G PAM
2. U4915: SPDT
3. U4912: LNA
4. U4913: SPDT

-  G850/G900 Tx
-  G850/G900 Rx common comp.
-  G850 Rx
-  G900 Rx



MB TRx Common Comp.

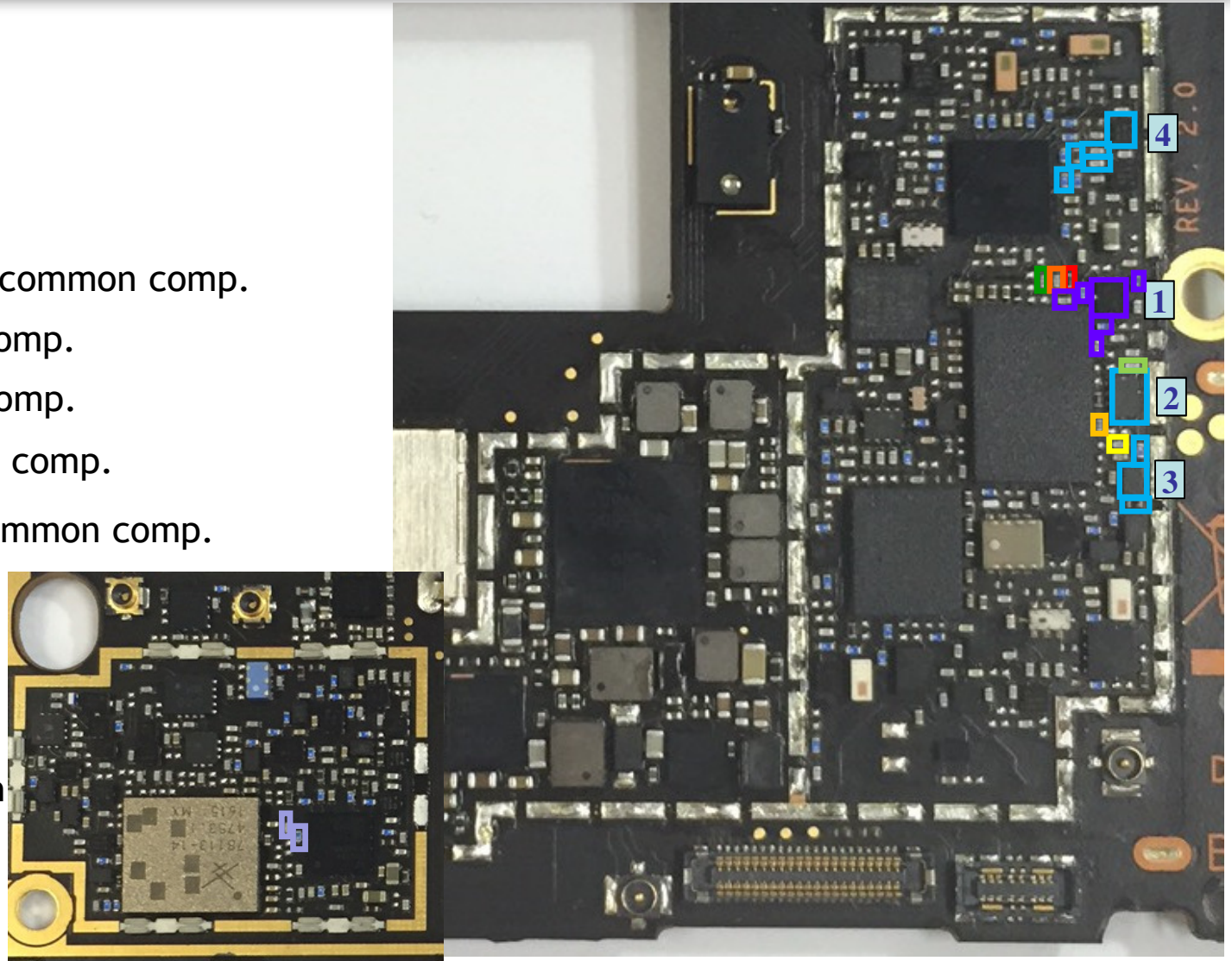
1. CON4701: Connector
2. U4708: DPDT
3. U4702: LPF
4. U4722: Coupler
5. U4809: Diplexer
6. U4806: MB FEM



MB TRx Comp.

1. U4804: Saw Filter
2. U4904: SP3T
3. U4910: LNA
4. U4909: SPDT

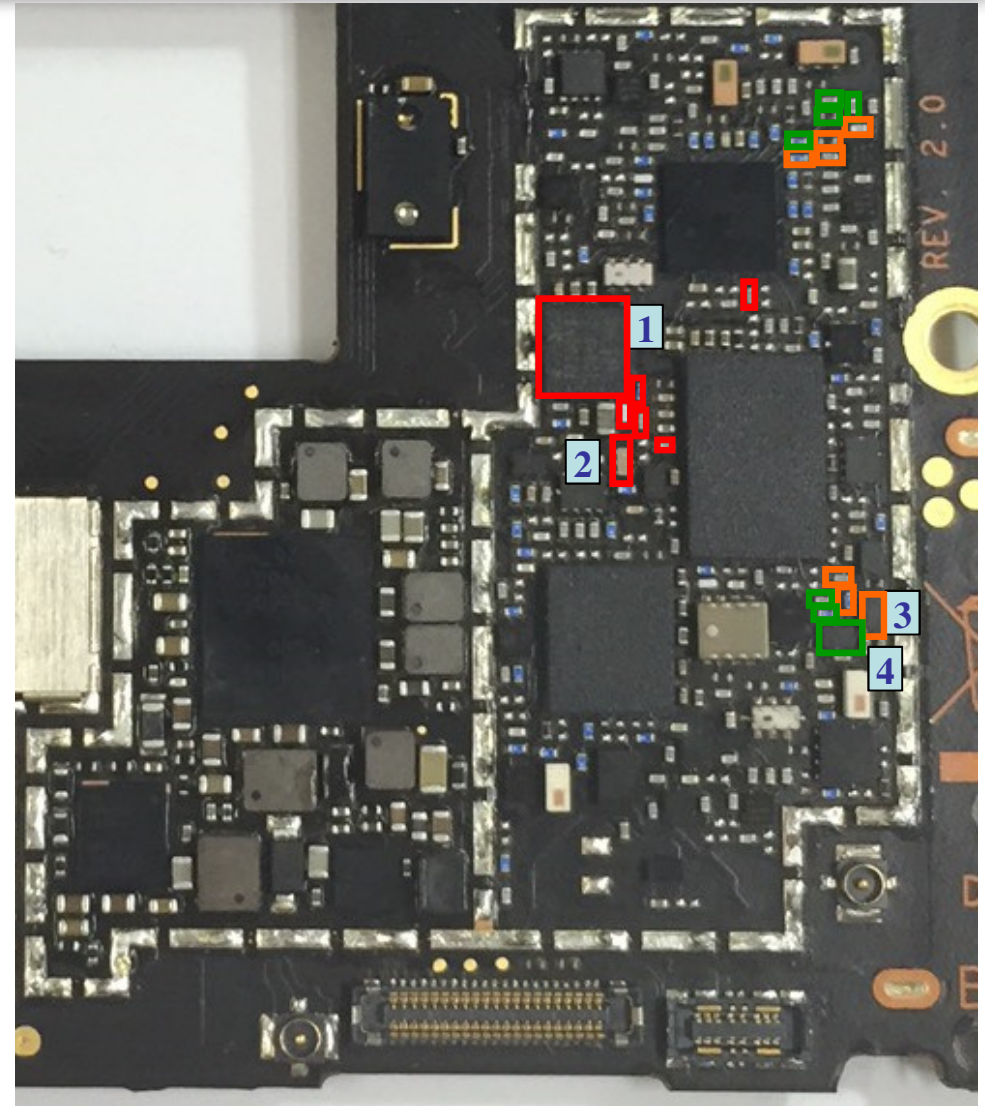
-  B1 B2 B3 B4 B39 Tx common comp.
-  B1 B2 Tx common comp.
-  B3 B4 Tx common comp.
-  B34 B39 Tx common comp.
-  B1 B4 B34 B39 Rx common comp.
-  B1 Rx comp.
-  B4 Rx comp.
-  B34 B39 Rx comp.
-  B1 Rx comp. CA Path



GSM DCS/PCS TRx (MB)

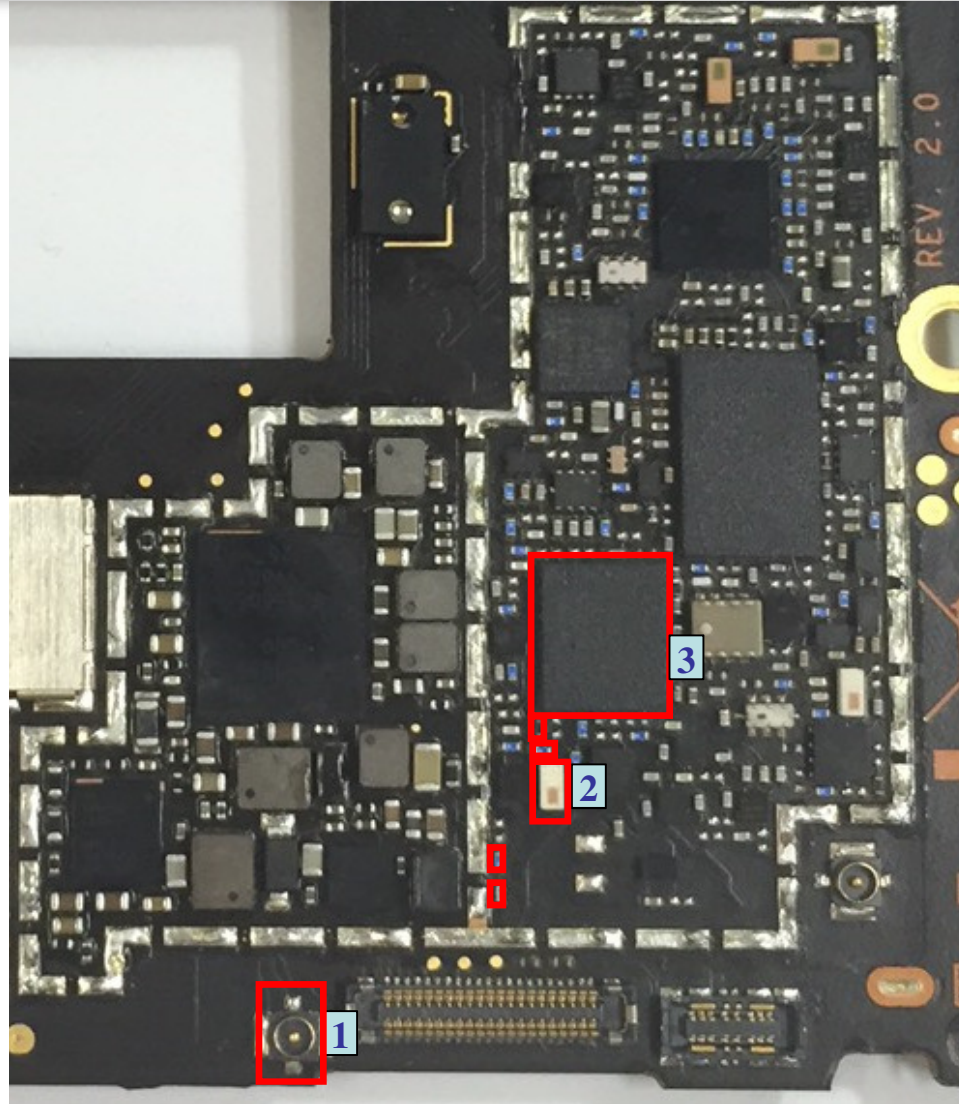
1. U4808: 2G PAM
2. U4807: LPF
3. U4908: LNA
4. U4920: LNA

-  DCS/PCS Tx
-  DCS Rx
-  PCS Rx



HB TRx Common Comp.

1. CON4704: Connector
2. U4703: LPF
3. U4801: FEM



HB TRx Comp.

1. U4820: SAW
2. U4802: SAW
3. U4803: SAW
4. U4919: SP3T
5. U4906: LNA
6. U4905: LNA
7. U4902: HPF

 B7 B30 B38 B40 B41 Tx common comp.

 B30 TRx comp.

 B40 TRx comp.

 B41 TRx comp.

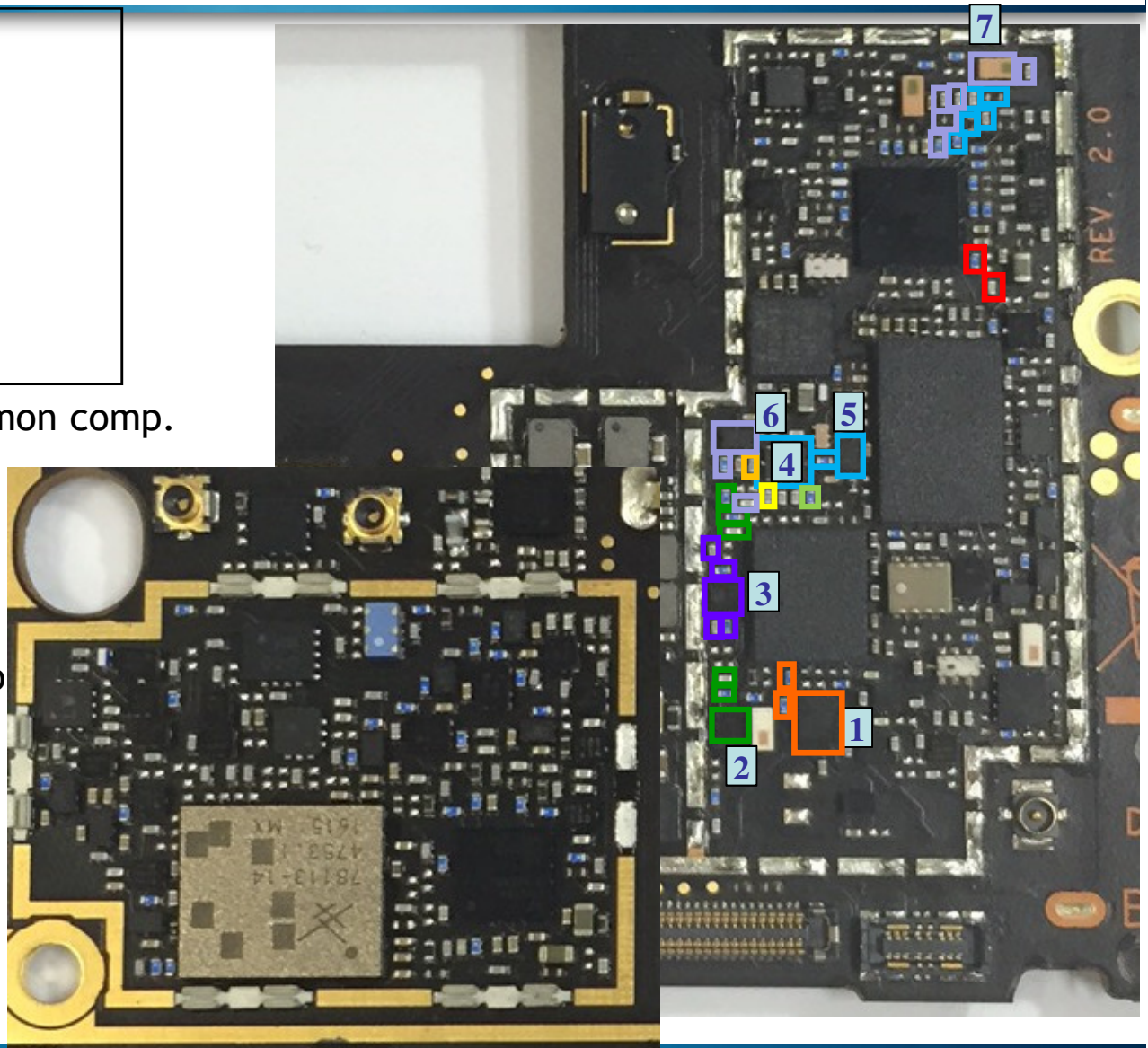
 B7 B30 B40 Rx common comp

 B7 Rx comp.

 B30 Rx comp.

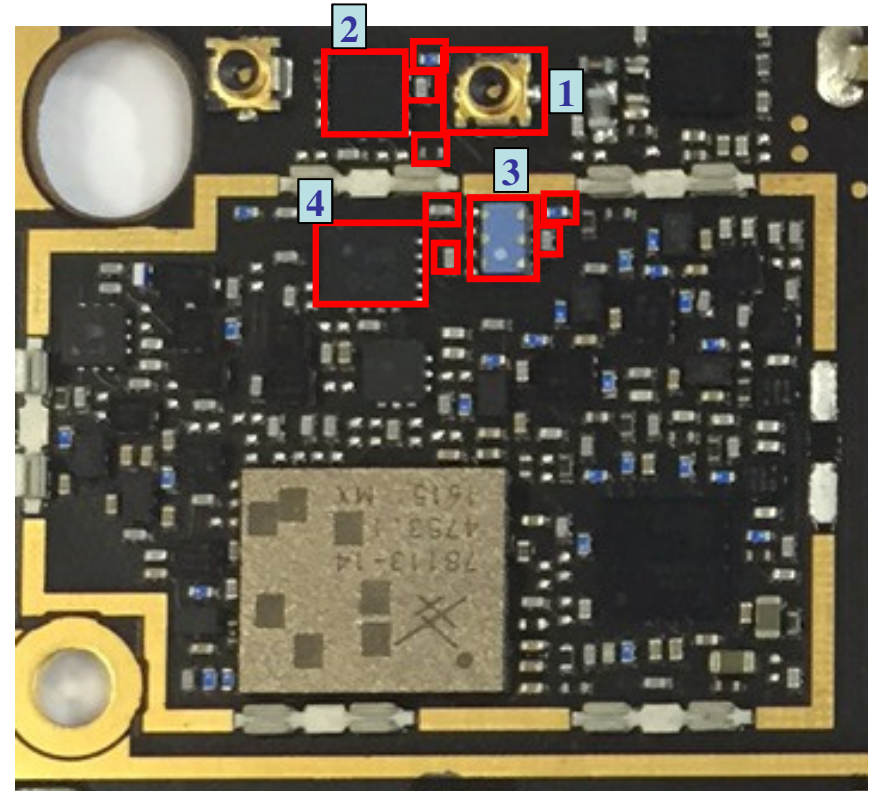
 B40 Rx comp.

 B38 B41 Rx common comp.



LB/MB Div Rx common comp

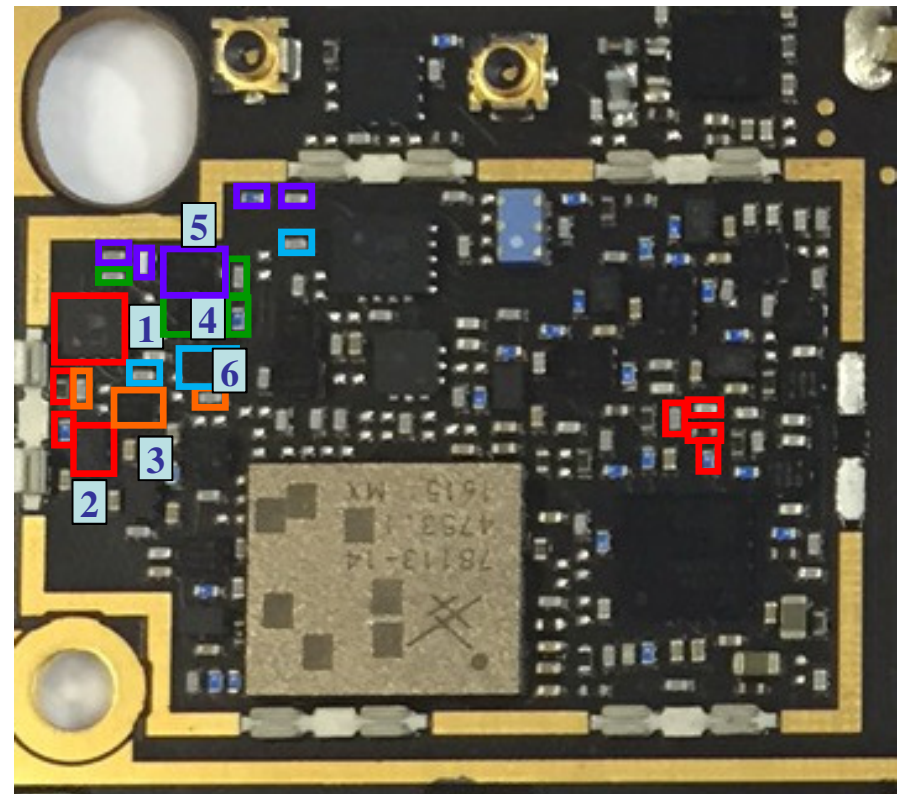
1. Con4702: connector
2. U4710: HPF
3. U4713: Diplexer
4. U4705: ASM



LB Div Rx

1. U5125: SP4T
2. U5126: LNA
3. U5123: Saw Filter
4. U5127: Saw Filter
5. U5128: Saw Filter
6. U5124: Saw Filter

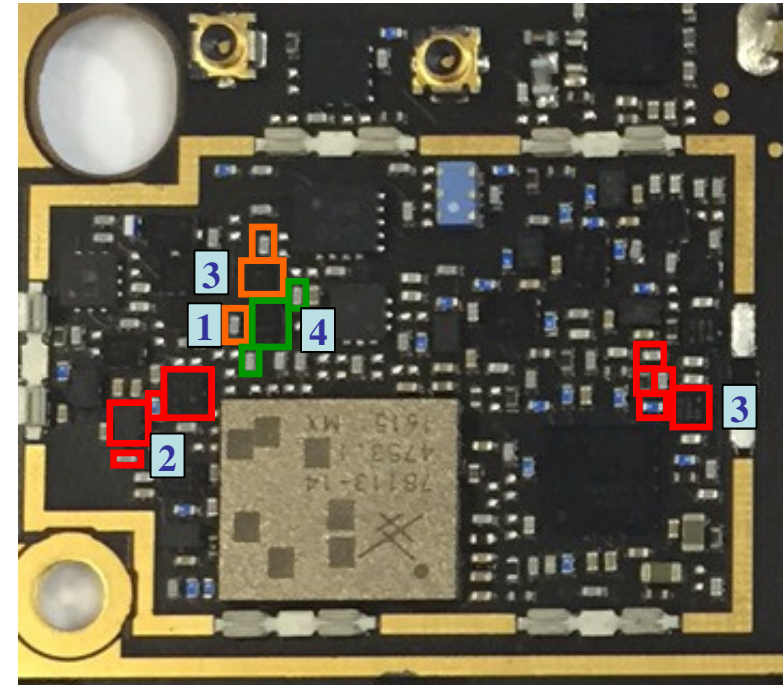
-  B12 B17 B20 B28 B29Rx common comp.
-  B12 B17 Rx common comp.
-  B20 Rx comp.
-  B28 Rx comp.
-  B29 Rx comp.



LB Div Rx(cont')

1. U5122: SPDT
2. U5118: LNA
3. U5117: Saw Filter
4. U5121: Saw Filter

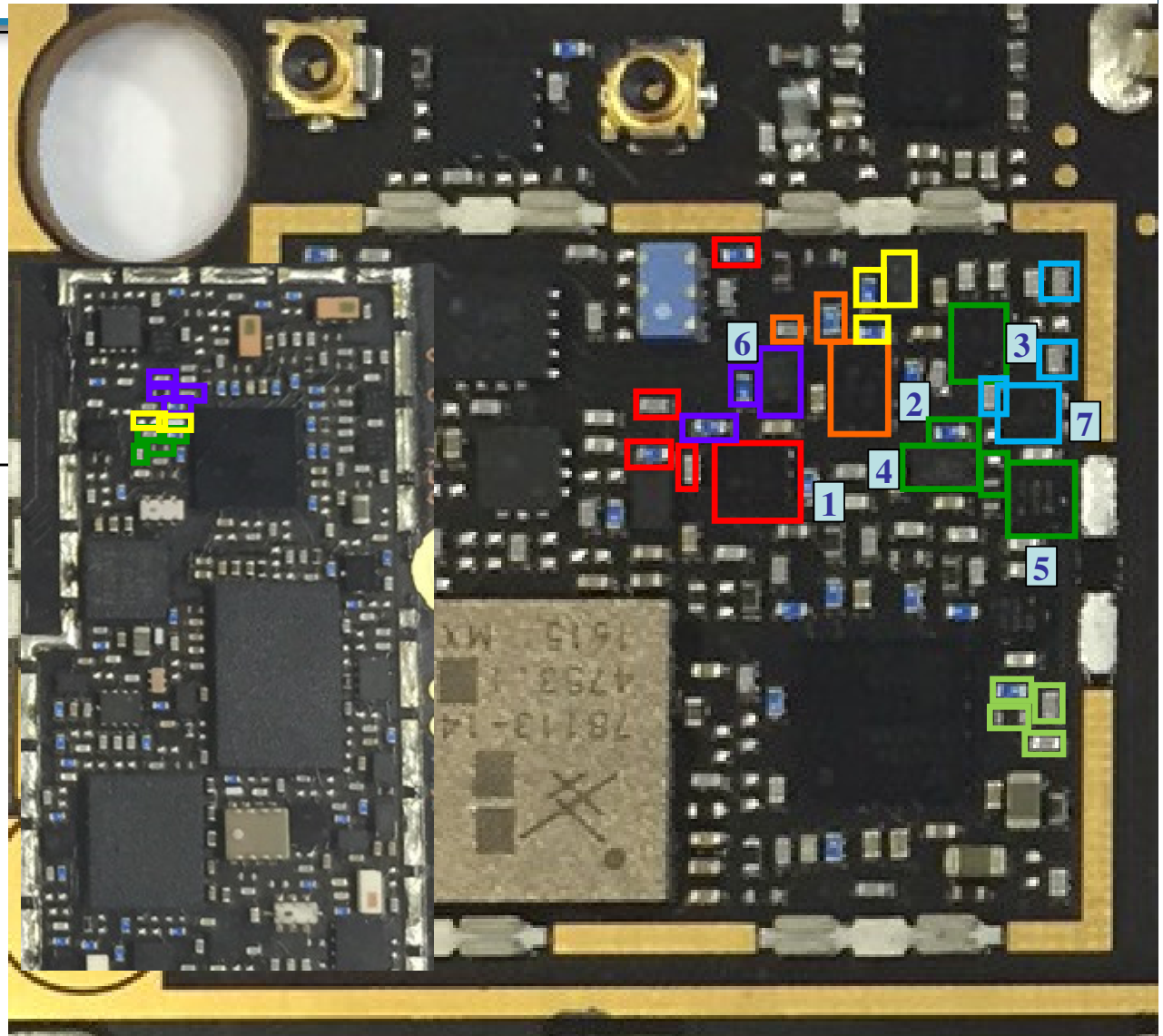
-  B5 B8 Rx common comp.
-  B8 Rx comp.
-  B5 Rx comp.



MB Div Rx

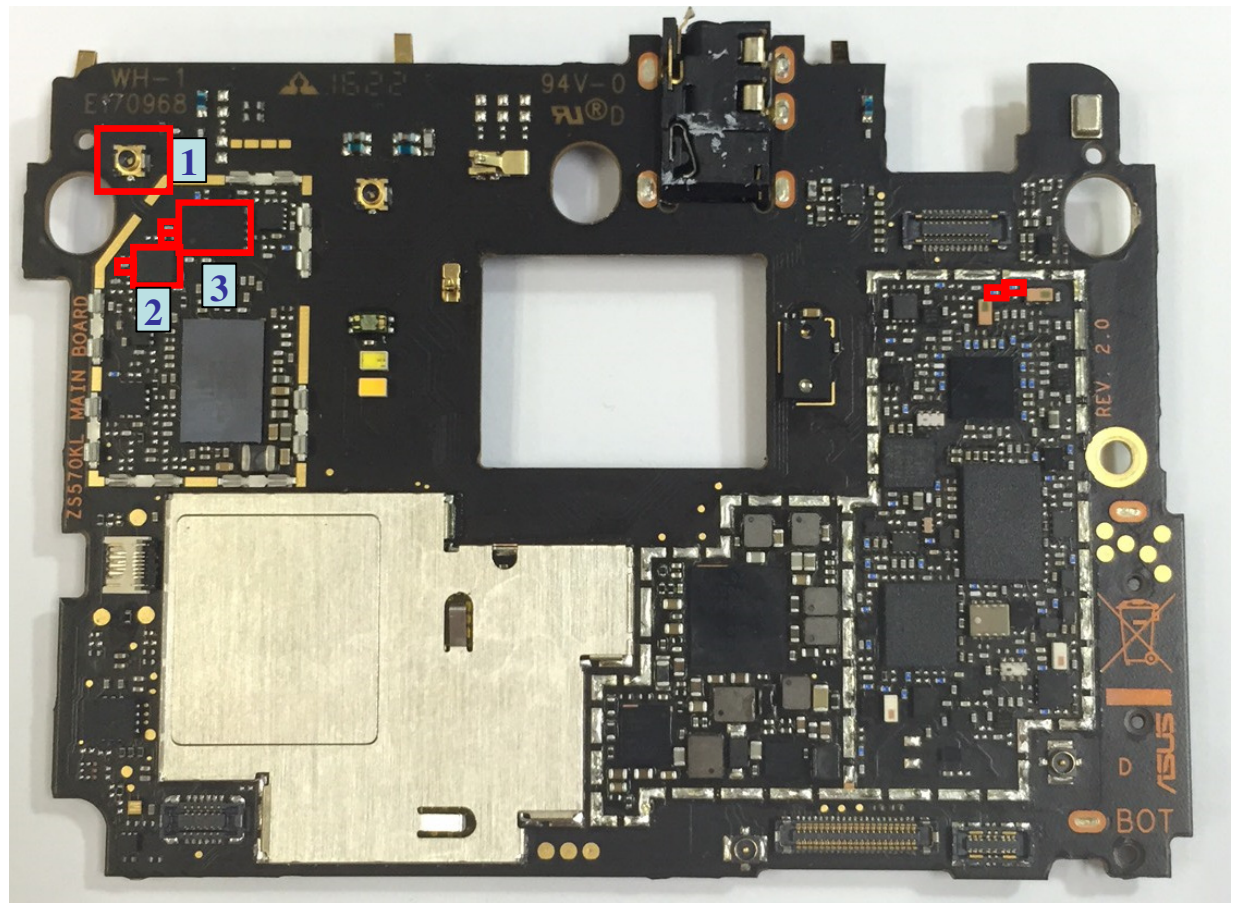
1. U5111: Saw Filter
2. U5115: Saw Filter
3. U5110: SP3T
4. U5113: LNA
5. U5114: SPDT
6. U5112: LNA
7. U5109: Saw Filter
8. U5116: LNA

-  B2 B4 Rx common comp.
-  B1 B3 Rx common comp.
-  B1 B4 B39 Rx common comp.
-  B2 Rx comp.
-  B39 Rx comp.
-  B3 Rx comp.
-  B1 Rx comp. CA path



HB Div Rx common comp

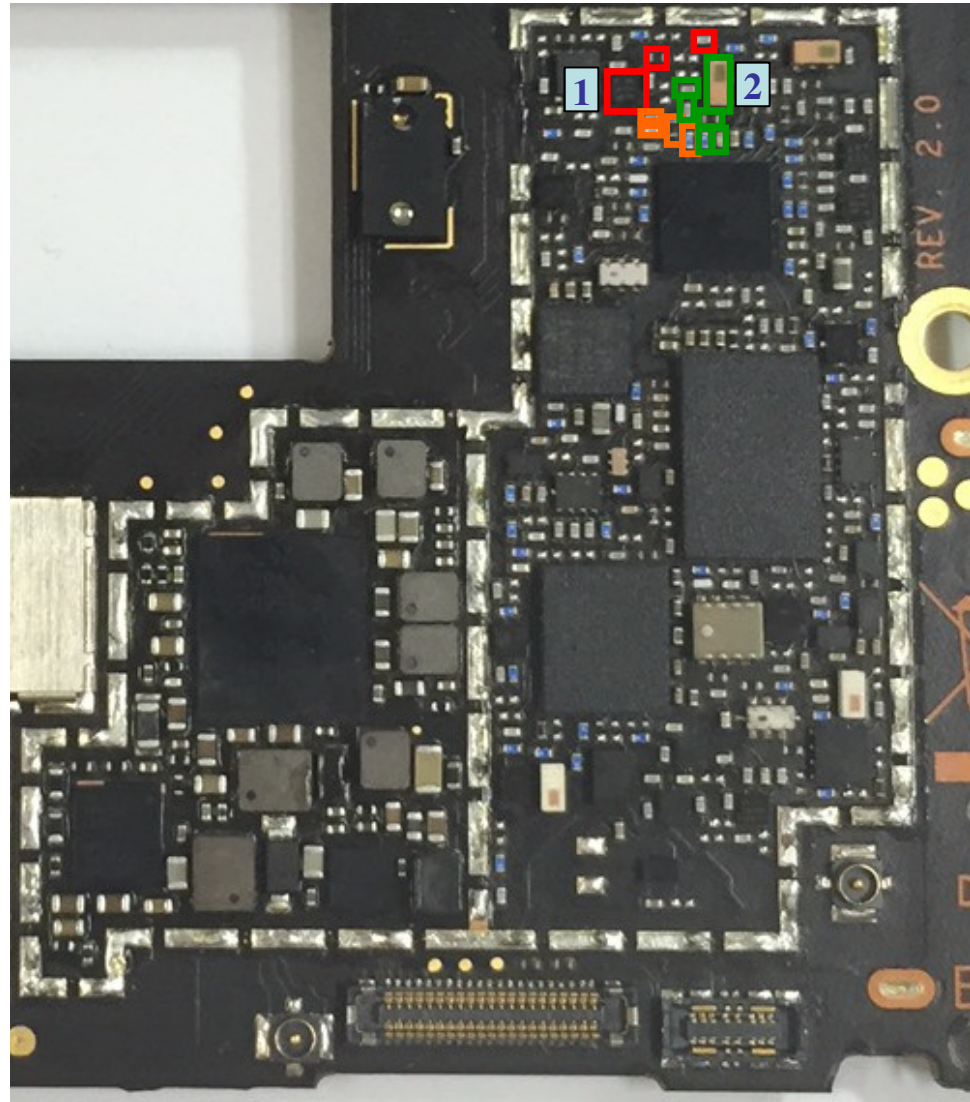
1. Con4702: connector
2. U4719: Extractor Filter



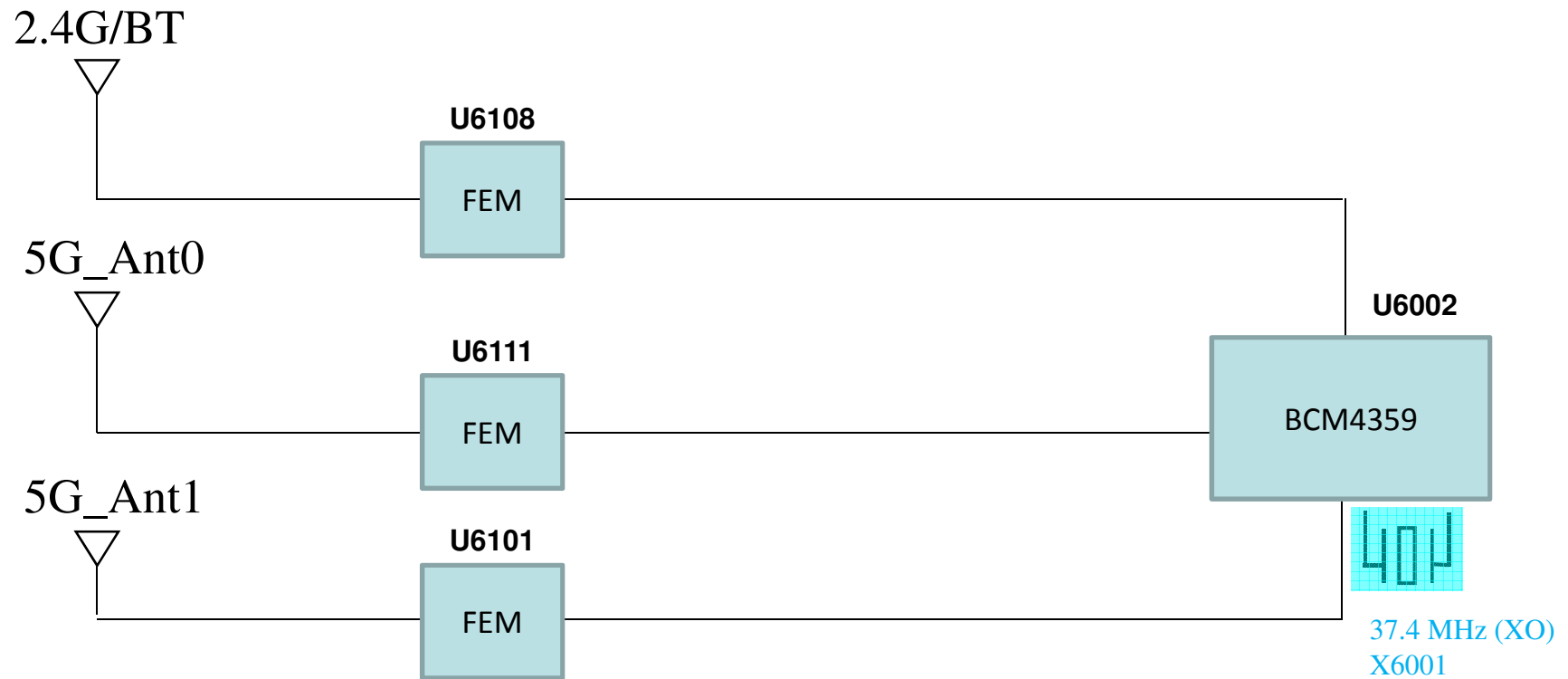
HB Div Rx

1. U5105: SPDT
2. U5132: LNA

- ☐ B7 B30 B38 B40 B41 Rx common comp.
- ☐ B7 B30 B40 Rx common comp.
- ☐ B38 B41 Rx common comp.



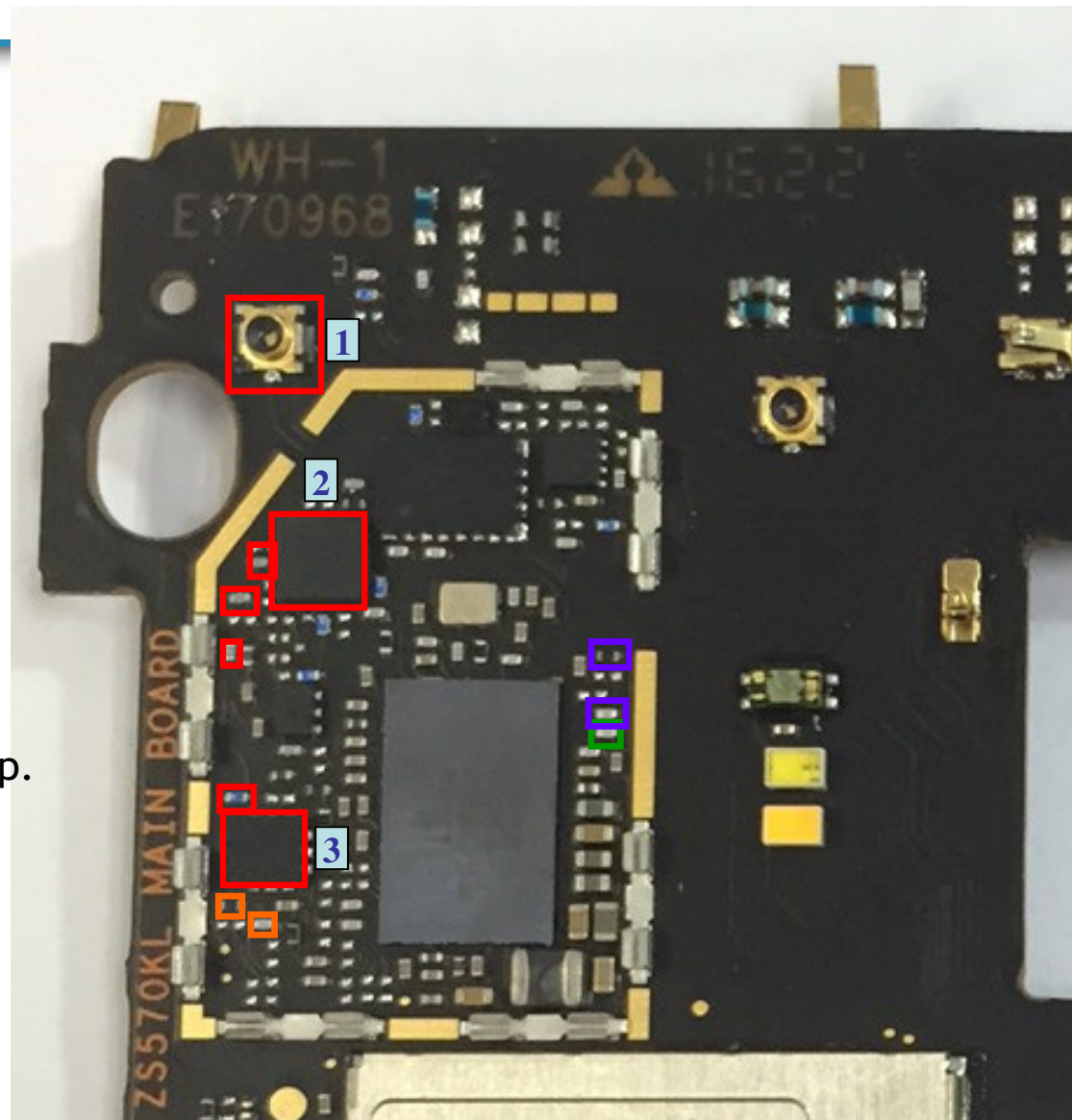
Wifi/BT Block Diagram



WiFi 2.4G/BT Comp.

1. CON4703: Connector
2. U4719: Extractor
3. U6108: FEM

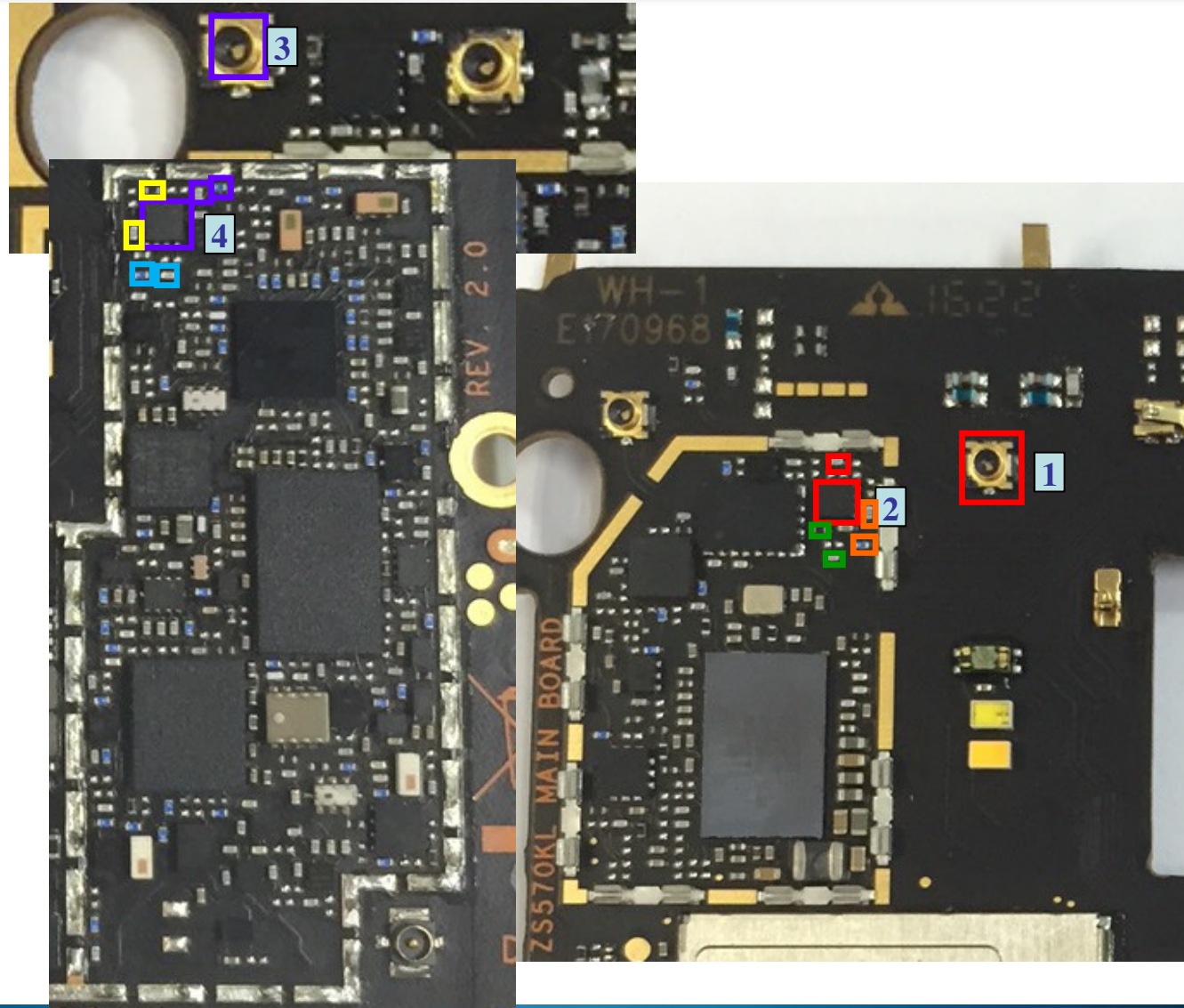
- WiFi 2.4G/BT TRx
- WiFi 2.4G TX
- BT Tx
- WiFi 2.4G BT Rx common comp.



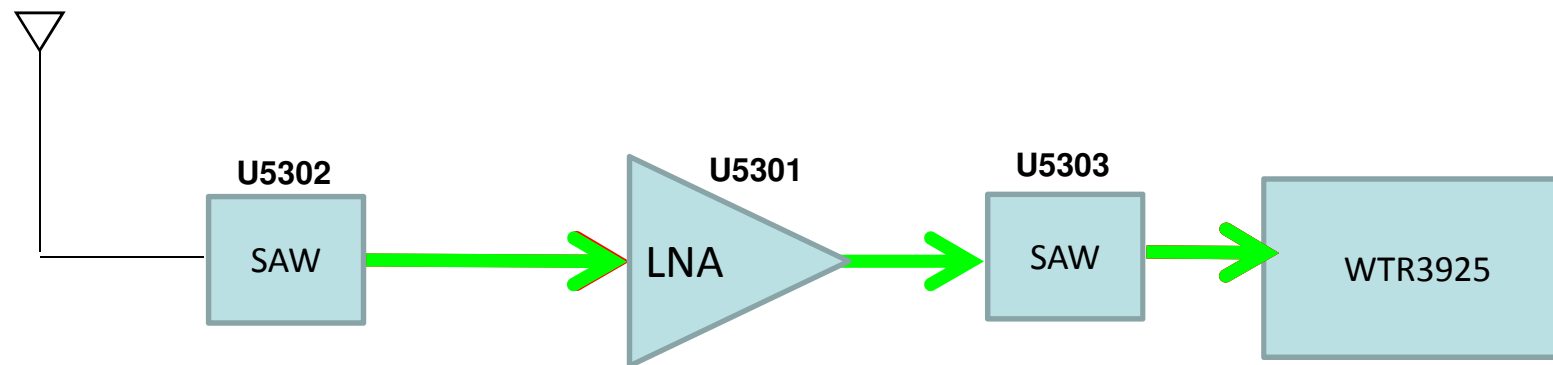
WiFi 5G Comp.

1. CON6101: Connector
2. U6101: FEM
3. Con6103: Connector
4. U6111: FEM

- WiFi 5G_Ant1 TRx
- WiFi 5G_Ant1 TX
- WiFi 5G_Ant1 RX
- WiFi 5G_Ant0 TRX
- WiFi 5G_Ant0 TX
- WiFi 5G_Ant0 RX

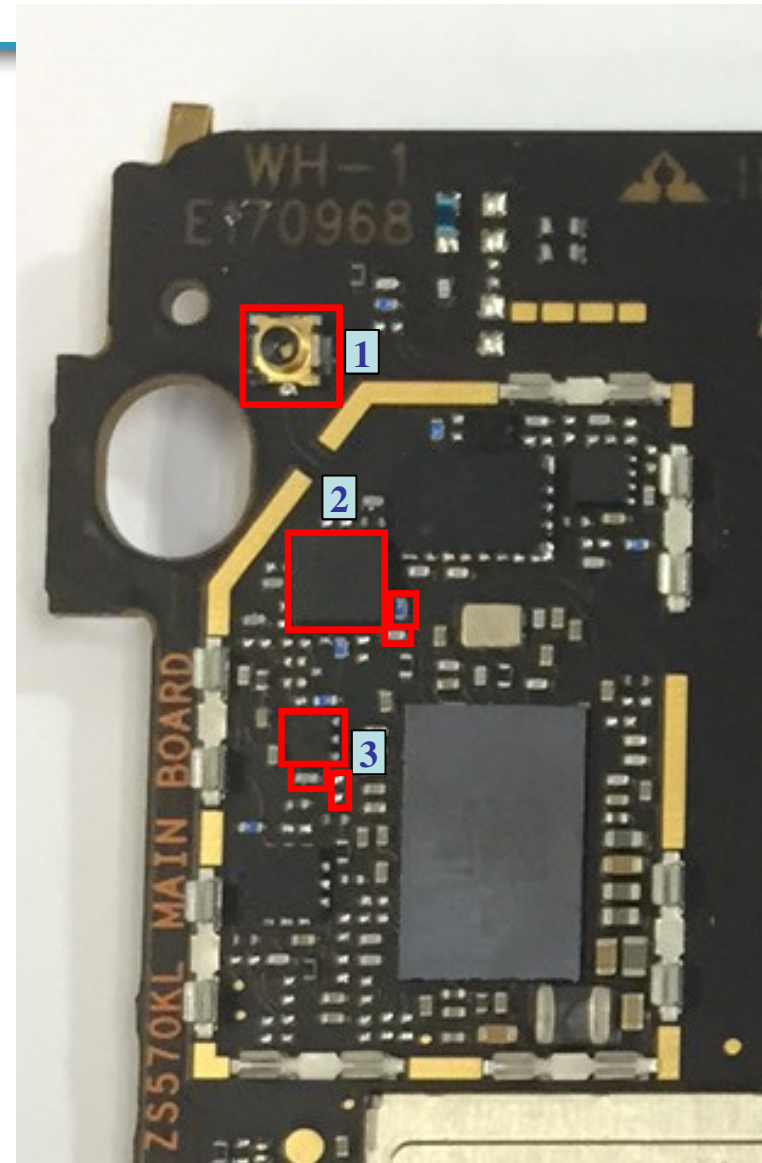
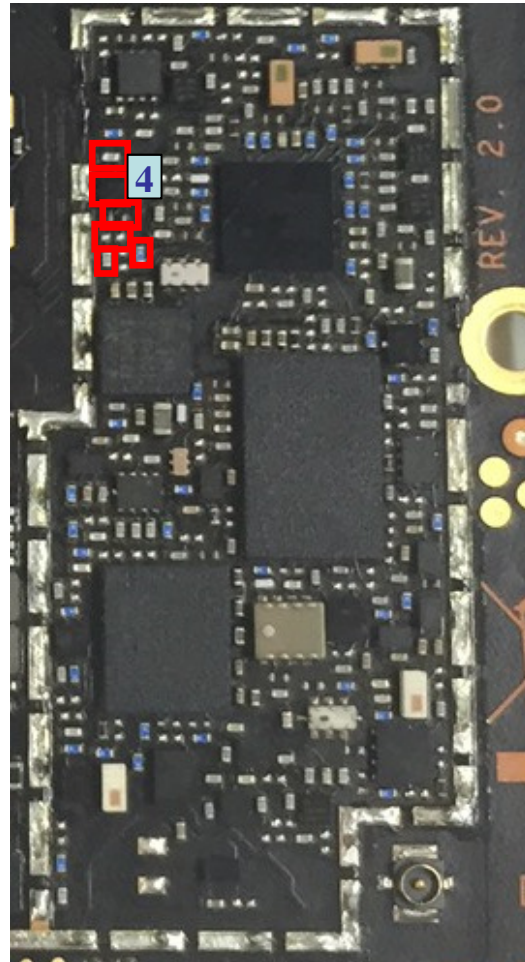


GPS Block Diagram



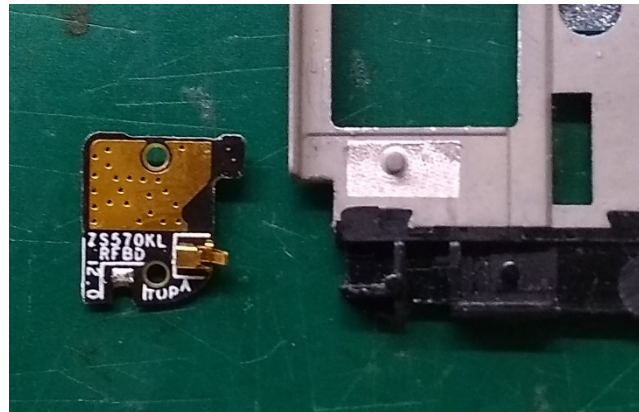
GPS Comp.

1. CON4703: Connector
2. U4719: Extractor
3. U5301: LNA
4. U5303: Saw Filter



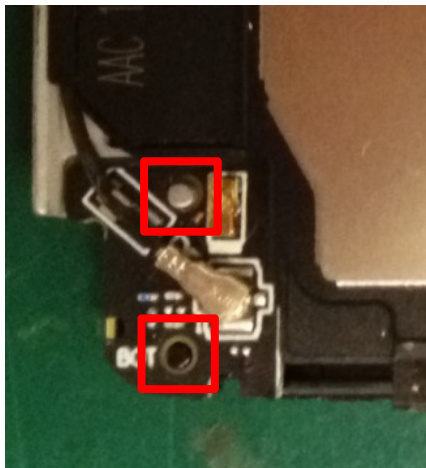
RF 小板組裝

- 組裝前，化金區與雷雕區使用清潔液擦拭
- 確認清潔液揮發後，貼上導電雙面膠，注意，手不可碰到導電雙面膠。

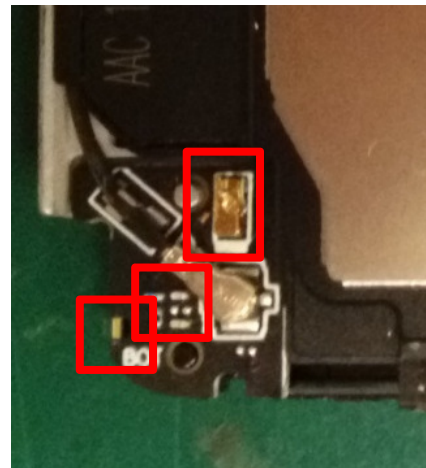


RF 小板組裝

- RF小板須確實貼附於正確位置，對位孔不可偏移，小板不可翹起。
- RF小板上元件有無損壞(電容、電感、電阻、彈片、接頭)，若有損壞，應立即更換。



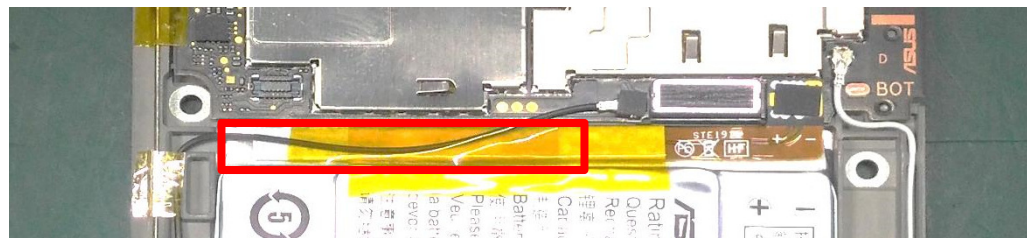
● 對位孔/柱



● 側頂彈片/電容電阻電感/上頂彈片

RF 小板組裝

- RF cable須確實放入理線槽，並且黏貼在規定區域



- 黑色cable一定走在紅色框電磁與主版空隙範圍，不可超過，並且使用膠帶固定
- 黑色cable一定走在線槽內，不可壓到

RF 小板組裝

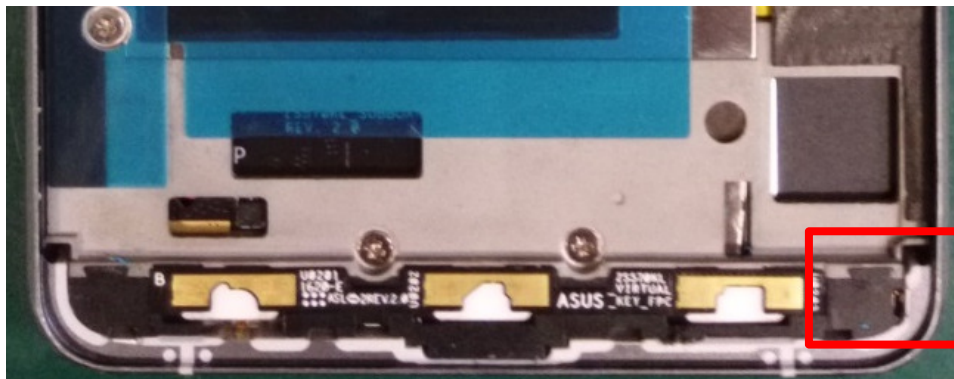
- 確認RF cable為IPX(目前立訊cable不在BOM表內)
- RF cable須確實扣好，輕撥不會彈起

- IPX cable上有IPX字樣，
且接頭顏色為銀色。立
訊則無字樣，接頭顏色
為黃色



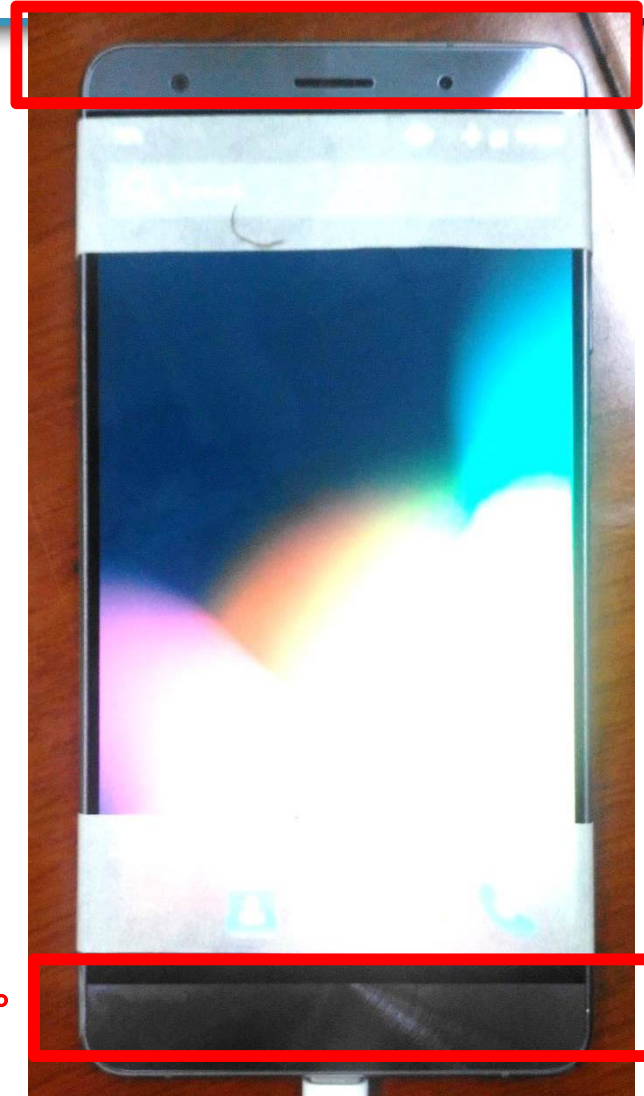
RF 小板組裝

- 確認彈片有無頂好，前框組裝良好，無不正常翹起



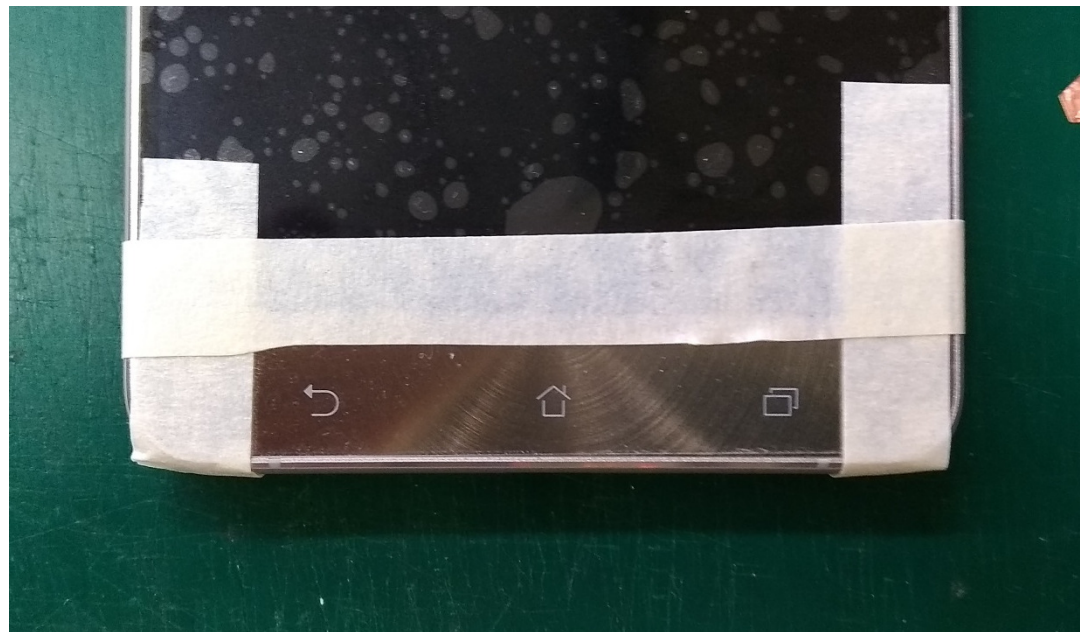
RF 小板組裝

- TP膠需貼附確實
 - TP組裝確實
-
- 螢幕扣上機台後，務必確認螢幕與機台密合度，上下不可有空隙，也不可以有位移，輕壓時不會有密合不良的喀喀聲。



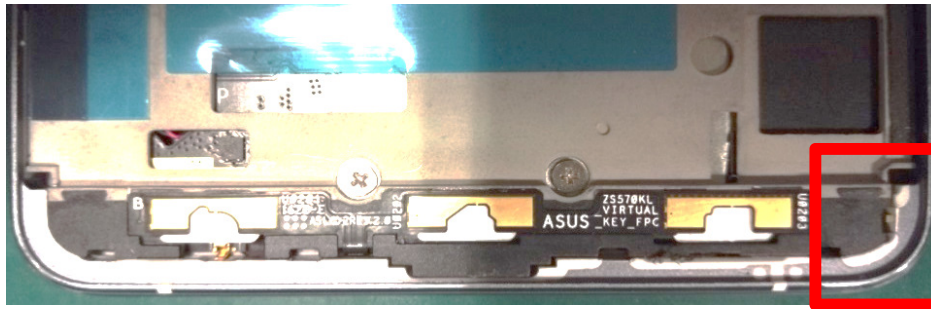
RF小板 High Band不良維修(B7/B40/B41)

- ERROR: BLER(1075) , Main Rx(1069)
- 檢查TP有無組裝確實
- 重新使用膠帶使TP固定，並且增加固定膠帶，使TP間隙較小。
- 若有相同ERROR CODE，重複上述步驟



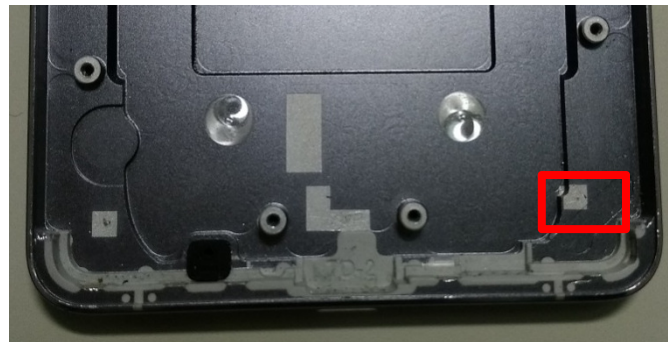
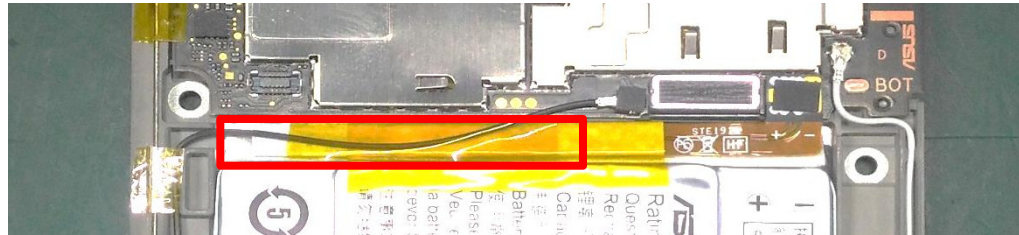
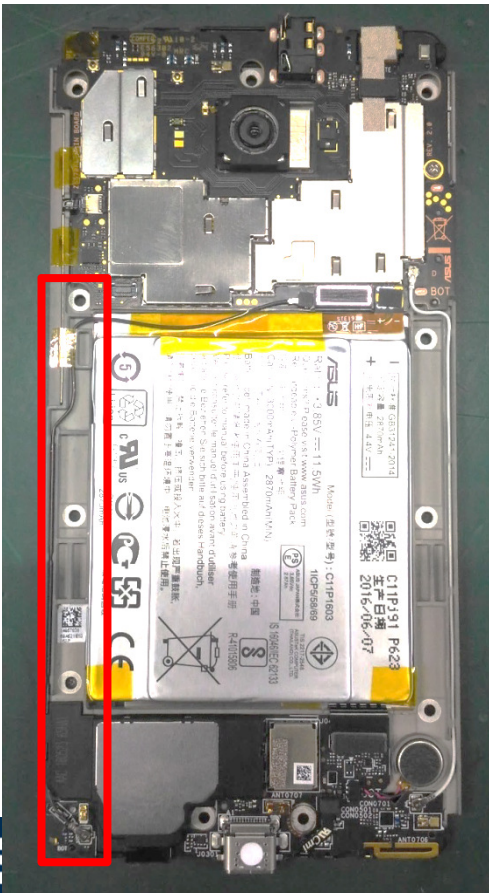
RF小板 High Band不良維修(B7/B40/B41)

- 拆下TP後，檢查TP膠有無確實黏貼，TP膠有無有無不正常凸起
- 若有凸起，有可能是cable未理好或是接頭未扣好
- TP膠撕起後，檢查測頂彈片有無頂好，有無歪斜



RF小板 High Band不良維修(B7/B40/B41)

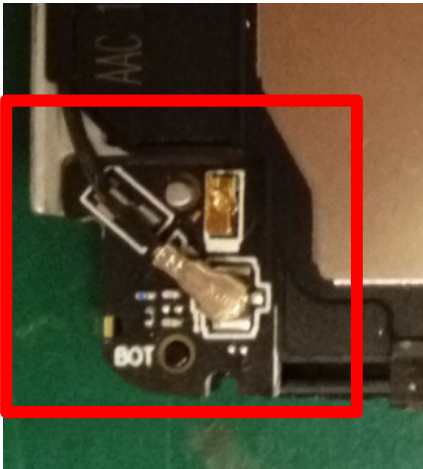
- 前框拆起後，檢查背蓋有無頂痕
- Cable外觀有無損壞，cable有無在指定區域，接頭有無扣好



- Cable在指定範圍內被蓋要有頂痕

RF小板 High Band不良維修(B7/B40/B41)

- RF小板有無翹起，元件有無損傷，彈片有無歪斜



ZS570 Agenda

➤ ZS570

- 2G 站
 - a. Main Antenna Low / Mid / High Band
 - b. Diversity Low / Mid / High Band
- Wi-Fi站
 - a. BT / Wi-Fi(2.4 GHz) / GPS
 - b. Wi-Fi(5 GHz)

ZS570



1. BT / Wi-Fi(2.4 GHz) / GPS / Diversity Ant. H Band
2. Diversity Ant. L/M Band
3. Main Ant. L/M Band
4. Main Ant. H Band

Low Band: 5 / 8 / 12 / 20 / 28

Mid Band: 1 / 2 / 3 / 4

High Band: 7 / 40 / 41

ZS570 Agenda – 2G站

➤ ZS570

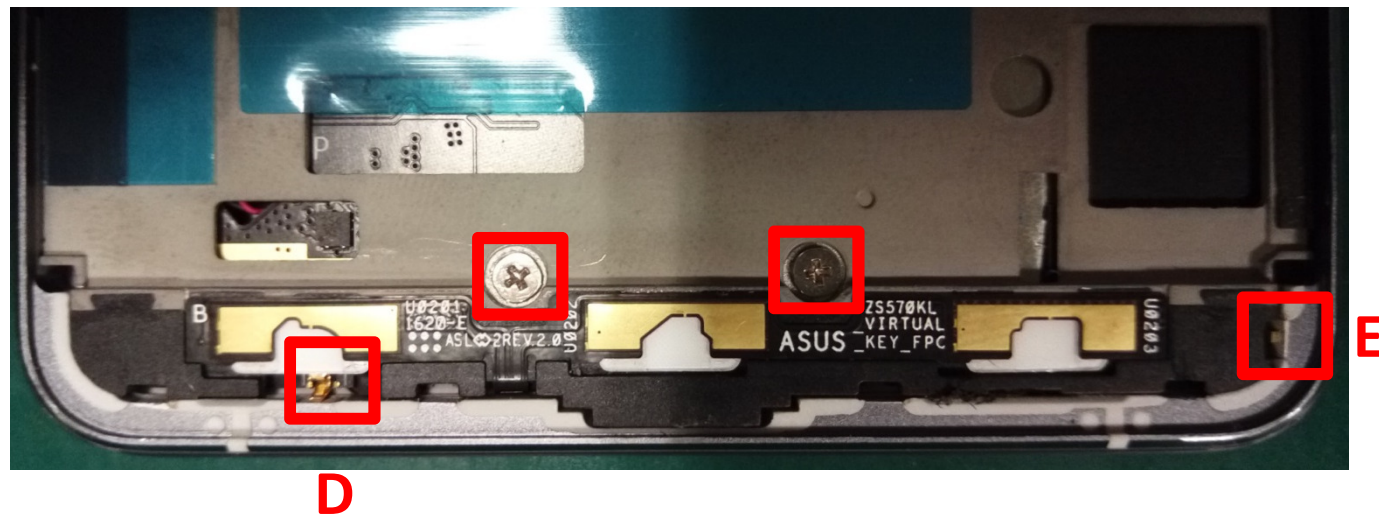
- 2G 站
 - a. Main Antenna Low / Mid / High Band
 - b. Diversity Low / Mid / High Band
- Wi-Fi站
 - a. BT / Wi-Fi(2.4 GHz) / GPS
 - b. Wi-Fi(5 GHz)

Main Antenna

1. 確認Touch Panel有貼好
2. 確認螺絲有鎖好
3. 確認彈片有頂到

若Main L/M Band Fail，可先確認彈片D

若Main H Band Fail，可先確認彈片E

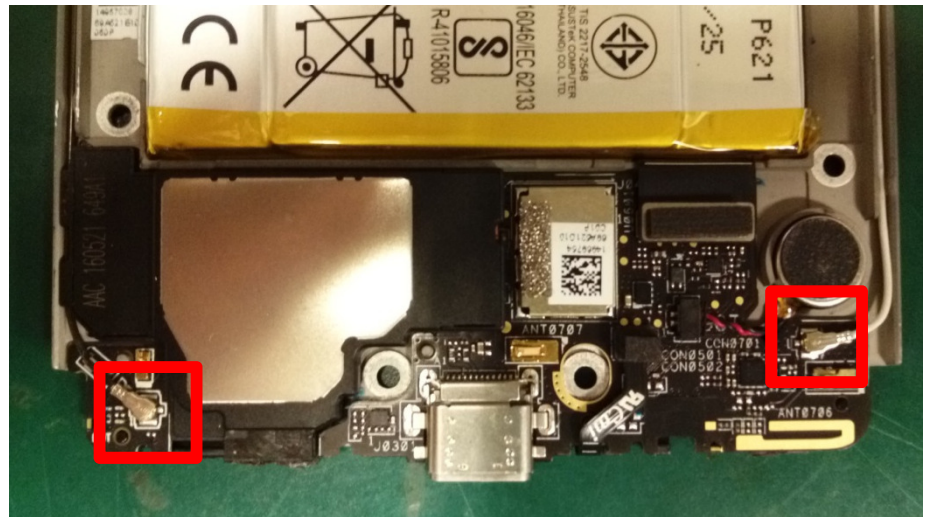


Main Antenna

3. 確認Cable有扣好或損壞

若Main L/M Band Fail，可先確認白色Cable

若Main H Band Fail，可先確認黑色Cable



Main Antenna

4. 更換小板複測

若Main L/M Band Fail，可更換小板A複測

若Main H Band Fail，可更換小板B複測

5. 更換背蓋複測

6. 更換主板複測



小板B

小板A

ZS570 Agenda – 2G站

➤ ZS570

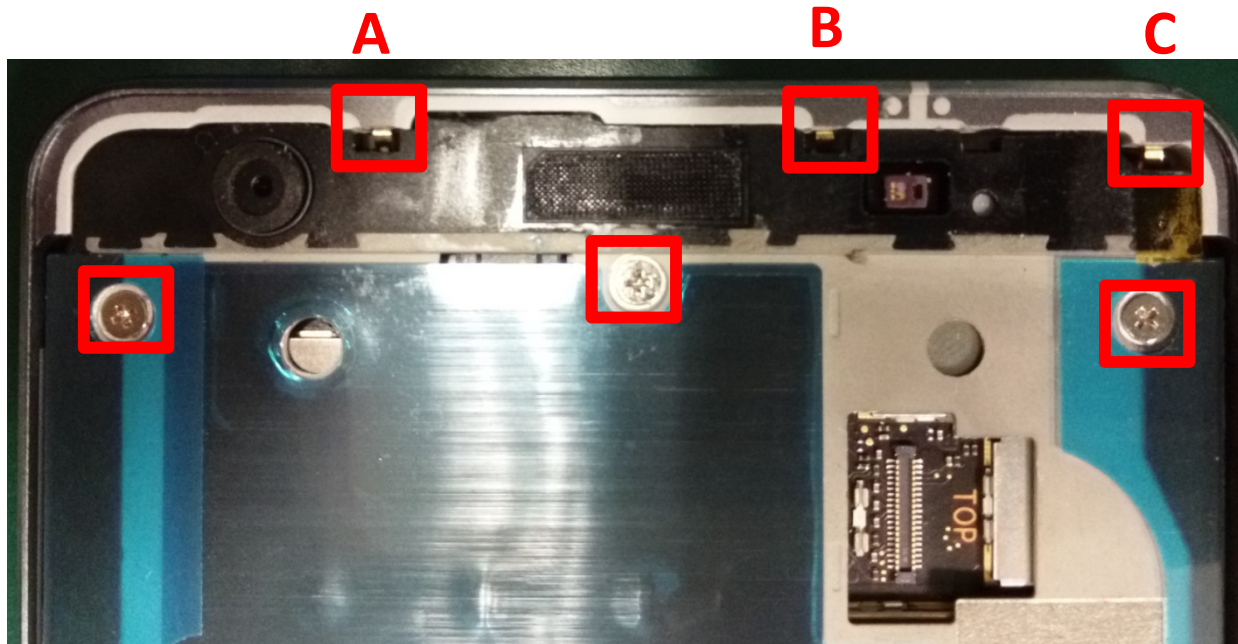
- 2G 站
 - a. Main Antenna Low / Mid / High Band
 - b. Diversity Low / Mid / High Band
- Wi-Fi站
 - a. BT / Wi-Fi(2.4 GHz) / GPS
 - b. Wi-Fi(5 GHz)

Diversity Antenna

1. 確認螺絲有鎖好
2. 確認彈片有頂好

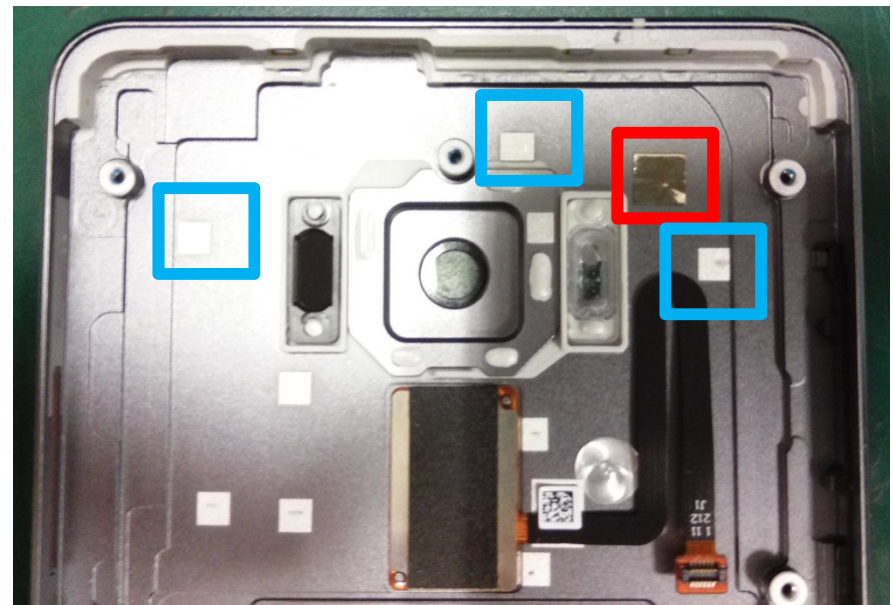
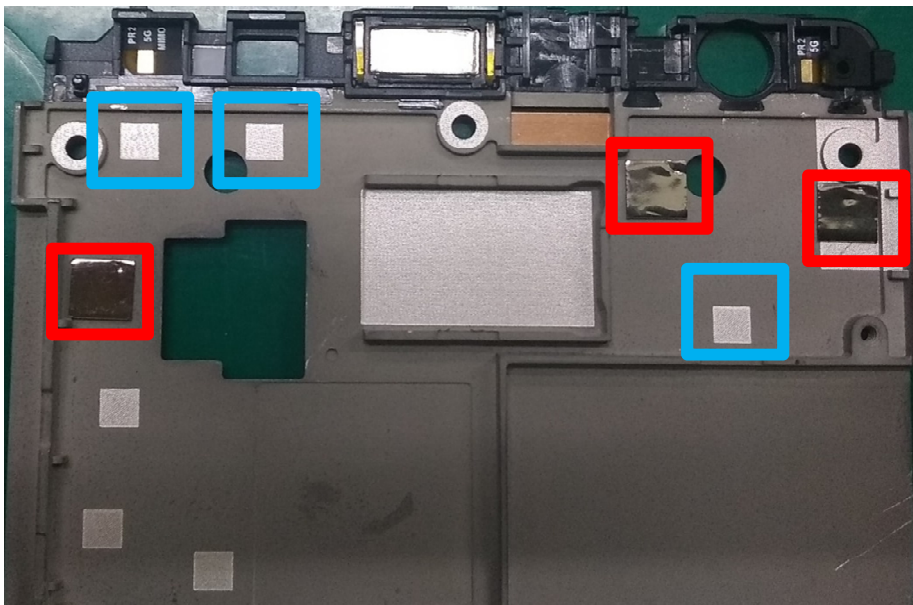
若 Div. Ant. L/M Band fail，可先確認彈片 A、B

若 Div. Ant. H Band fail，可先確認彈片 C



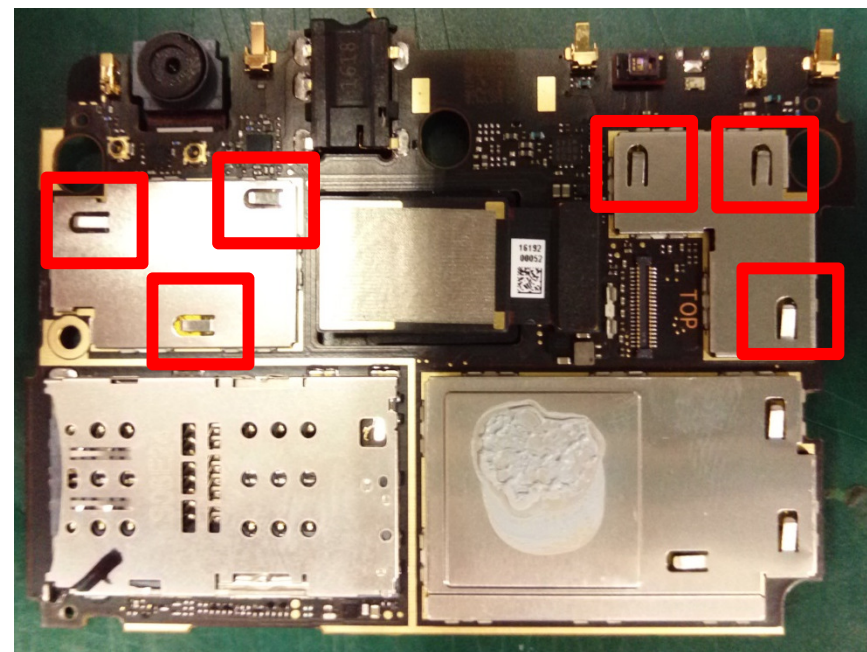
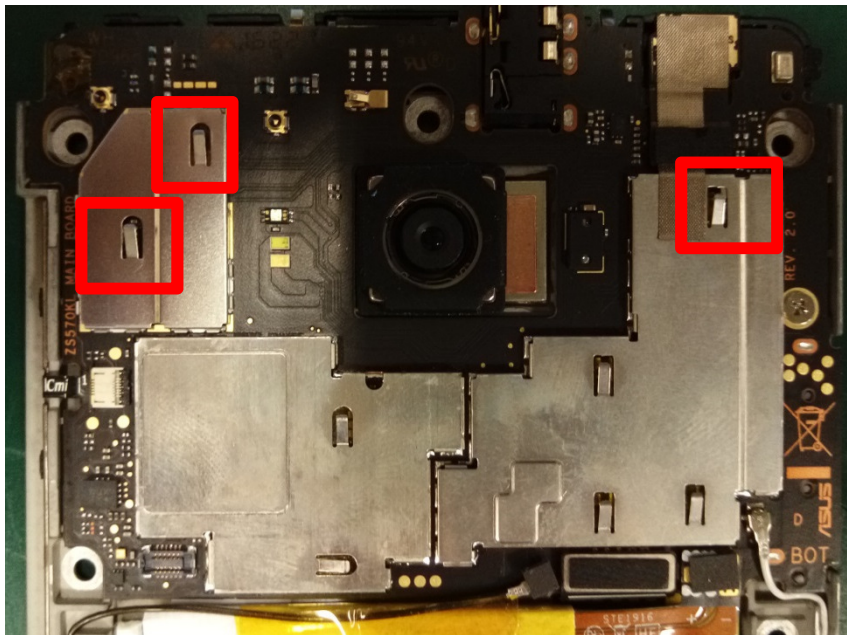
Diversity Antenna

3. 確認導電布有貼好
4. 確認雷雕點、導電布上有頂痕



Diversity Antenna

5. 將Shielding上的彈片撥高後複測
6. 更換背蓋複測
7. 更換主板複測



ZS570 Agenda – Wi-Fi站

➤ ZS570

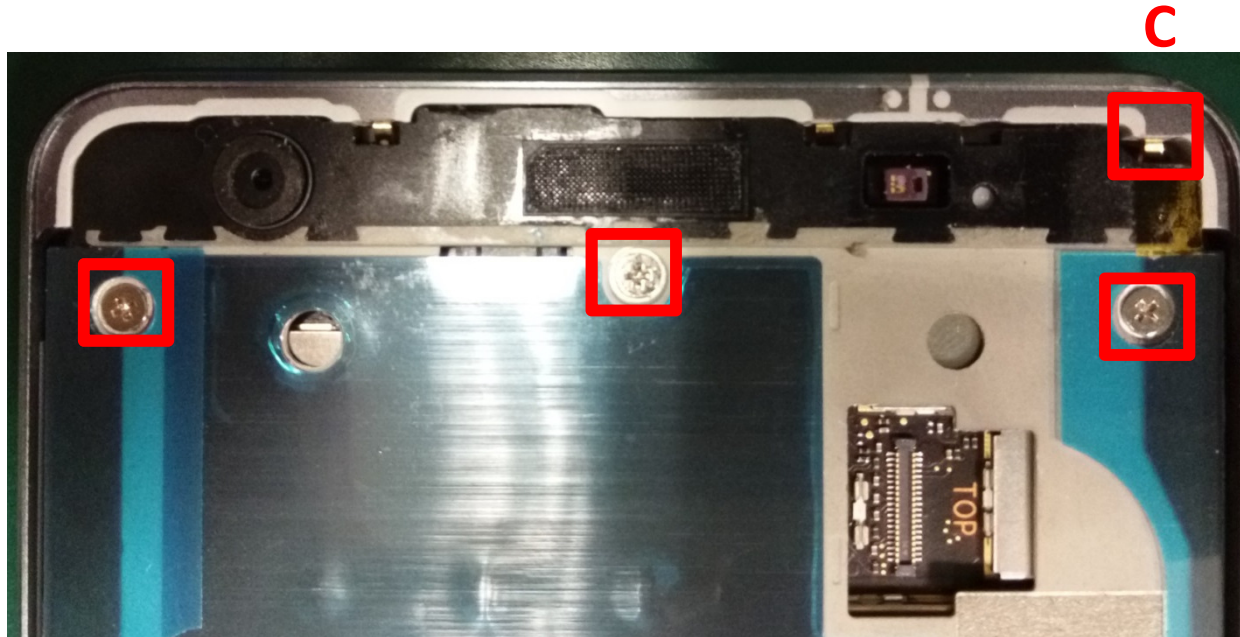
- 2G 站
 - a. Main Antenna Low / Mid / High Band
 - b. Diversity Low / Mid / High Band
- Wi-Fi站
 - a. BT / Wi-Fi(2.4 GHz) / GPS
 - b. Wi-Fi(5 GHz)

BT / Wi-Fi(2.4 GHz) / GPS

1. 確認螺絲有鎖好

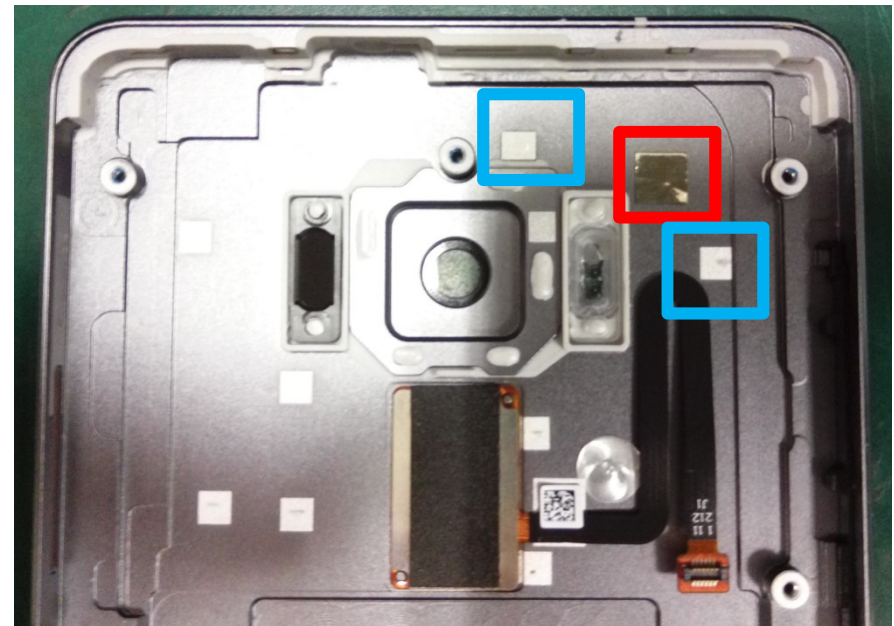
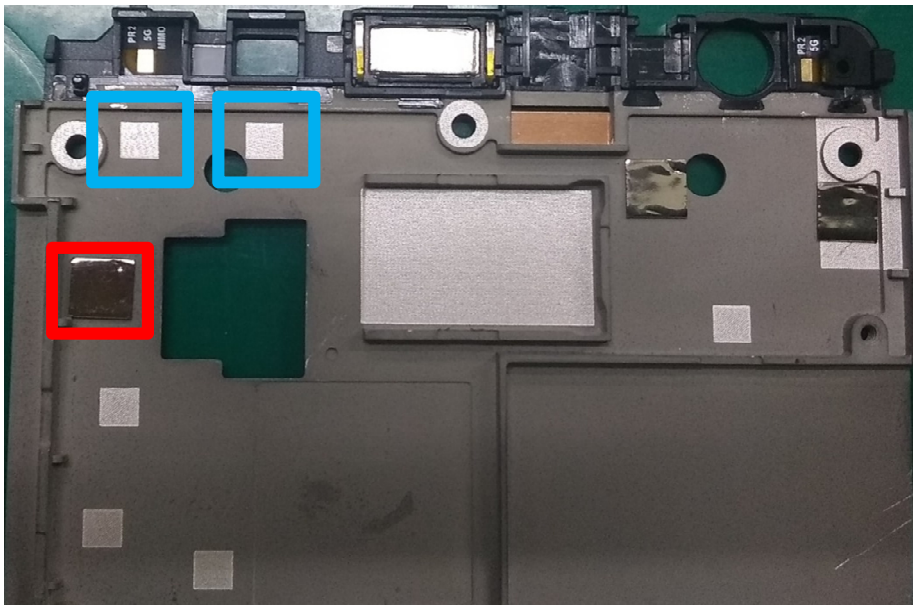
2. 確認彈片有頂到

若BT、Wi-Fi(2.4 GHz) Tx Power fail，可先確認彈片C



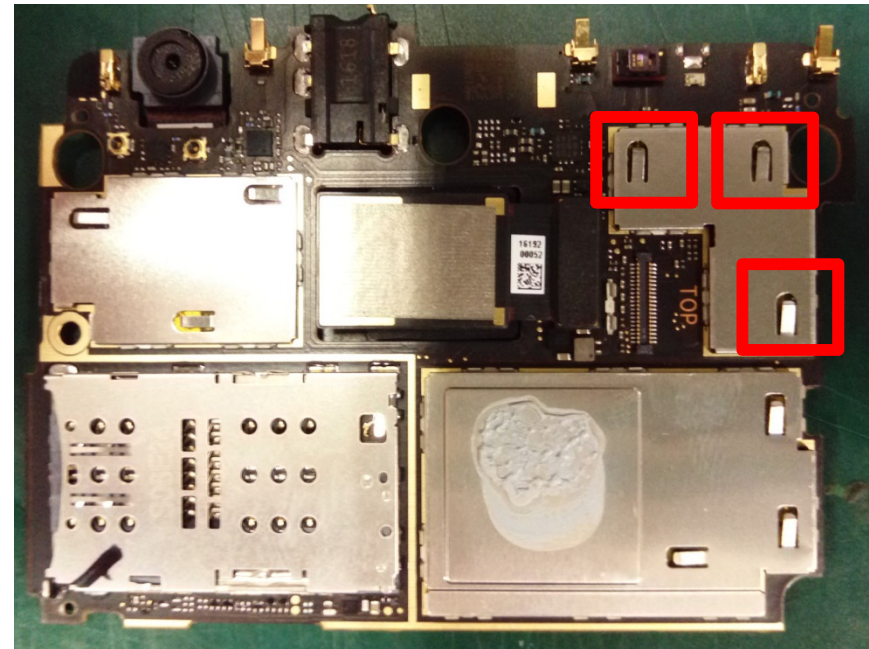
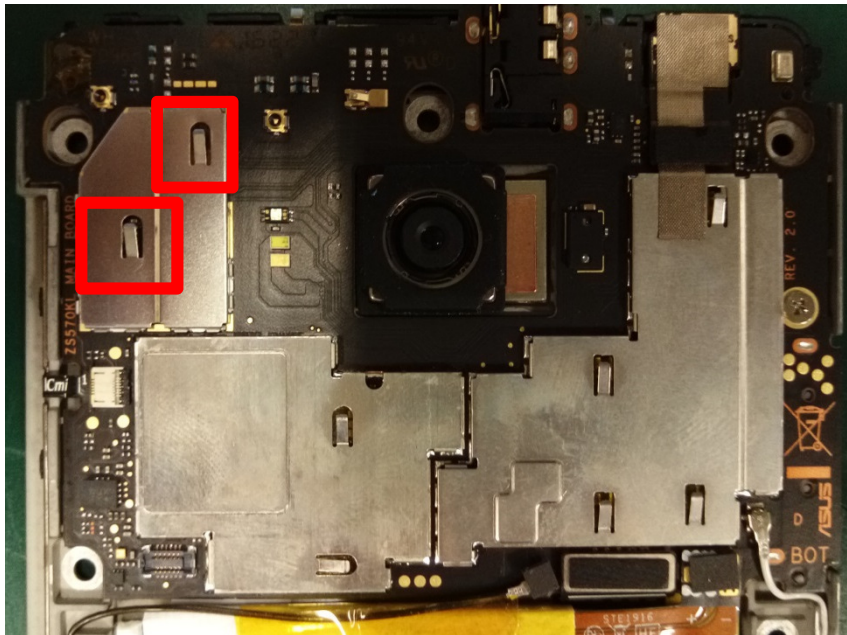
BT / Wi-Fi(2.4 GHz) / GPS

3. 確認導電布有貼好
4. 確認雷雕點、導電布有頂痕



BT / Wi-Fi(2.4 GHz) / GPS

5. 將彈片撥高後複測
6. 更換背蓋複測
7. 更換主板複測



ZS570 Agenda – Wi-Fi站

➤ ZS570

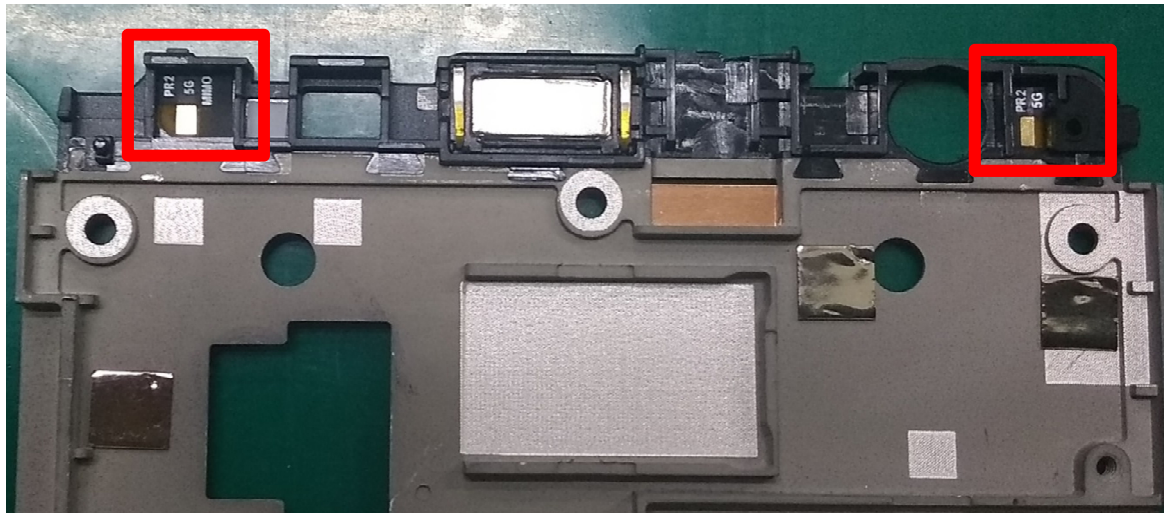
- 2G 站
 - a. Main Antenna Low / Mid / High Band
 - b. Diversity Low / Mid / High Band
- Wi-Fi站
 - a. BT / Wi-Fi(2.4 GHz) / GPS
 - b. Wi-Fi(5 GHz)

Wi-Fi (5 GHz)

1. 確認天線FPC有頂痕

11AN Aux

11AN Main

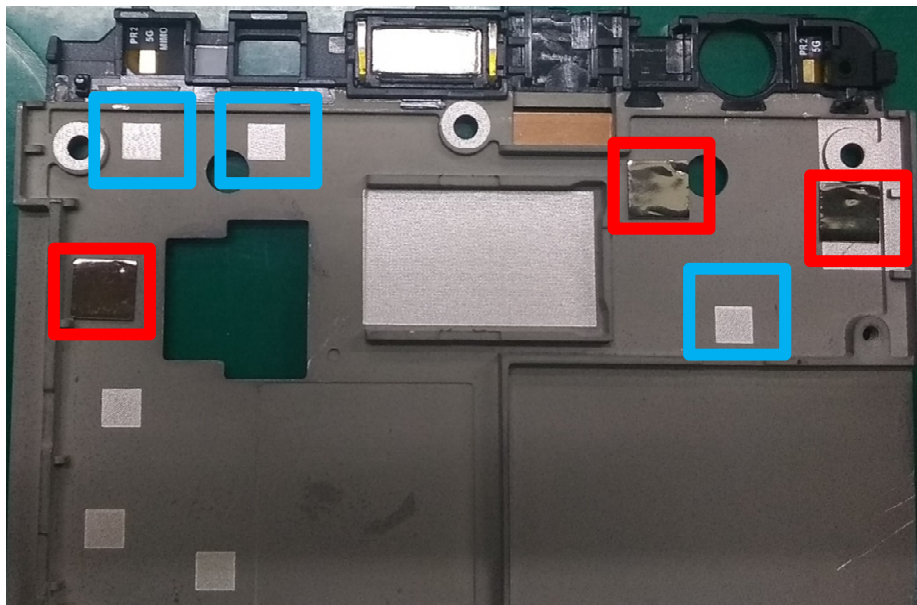


Wi-Fi (5 GHz)

2. 確認導電布有貼好
3. 確認雷雕點、導電布有頂痕

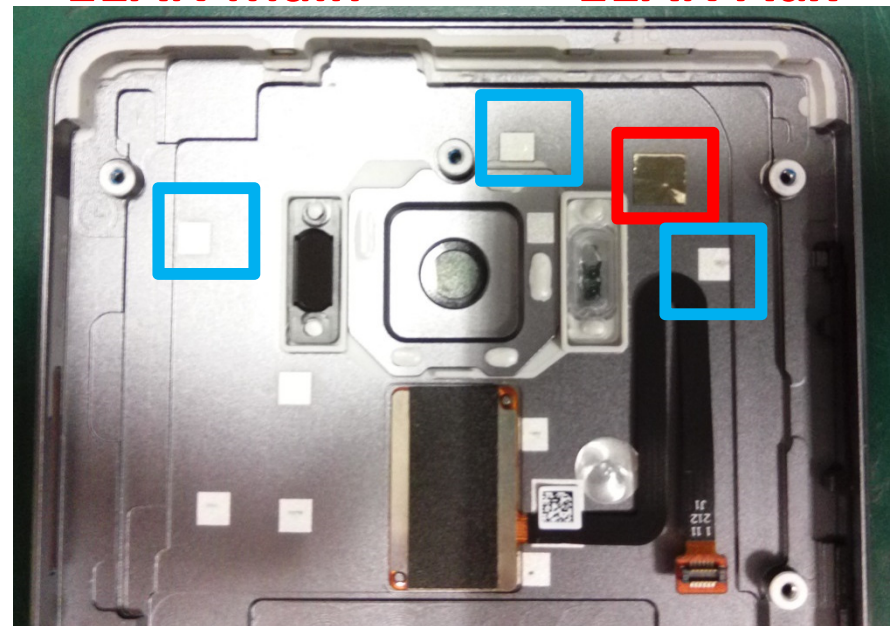
11AN Aux

11AN Main



11AN Main

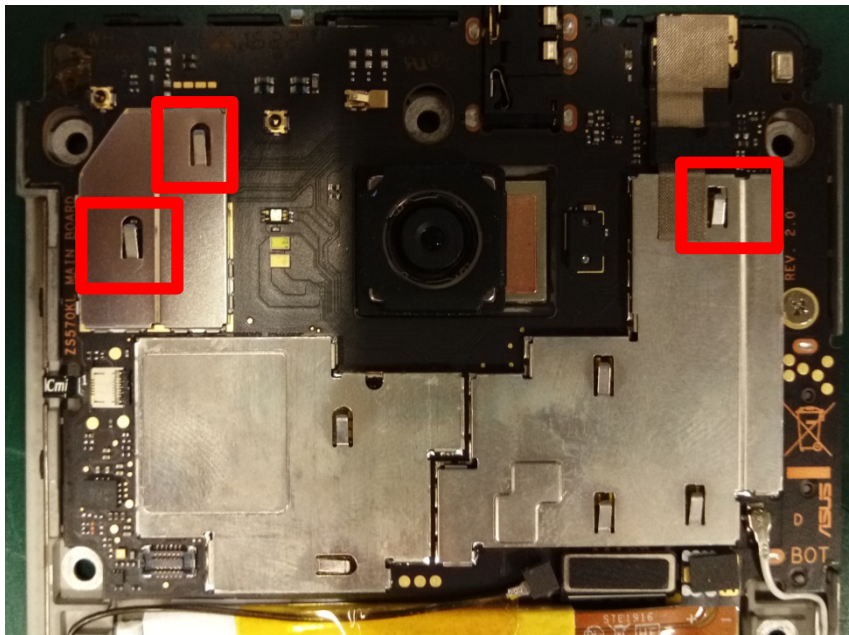
11AN Aux



Wi-Fi (5 GHz)

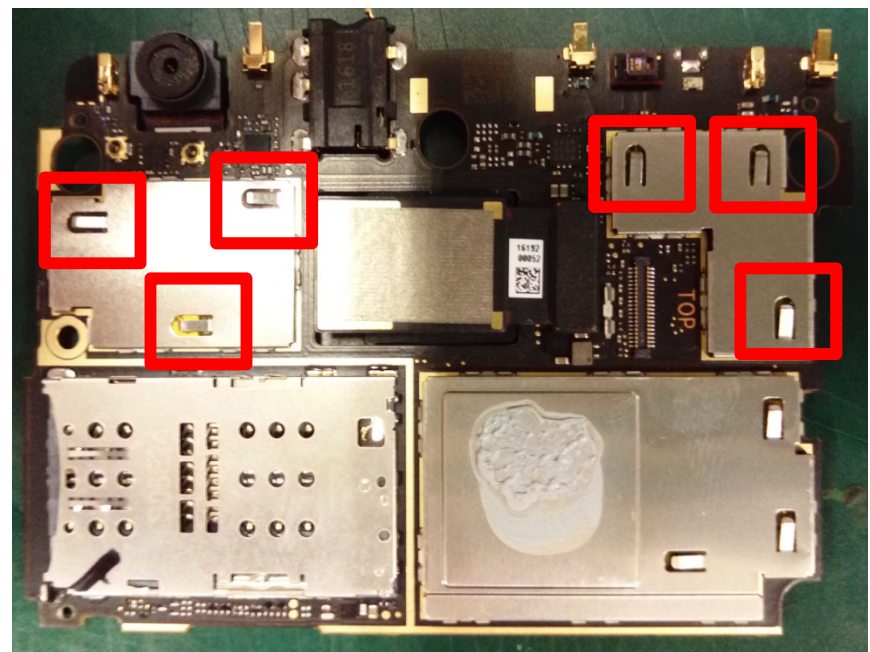
4. 將彈片撥高後複測

11AN Aux



11AN Main

11AN Main



11AN Aux

Note.

2G站主要Error:

- **1060: Main Antenna Tx Power Fail**
- **1069: Main Antenna Rx Fail**
- **1068: Diversity Antenna Rx Fail**
- **1062: EVM fail**
- **1067 、 1075: BLER fail**

Note.

Wi-Fi 站主要Error:

- **8085: BT Tx power 抓不到值**
- **8072: BT Tx power fail**
- **1108: Wi-Fi(5 GHz) Tx power 抓不到值**
- **1109: Wi-Fi(5 GHz) Tx power fail**
- **1111: Wi-Fi(5 GHz) EVM fail**
- **1127: Wi-Fi(5 GHz) Rx PER fail**

Thanks for your attention!